Western MA Per Pupil Expenditure Report

With selected districts for comparison

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https://github.com/timshores/schools/blob/main/ppe_report/output/WMPPE%20Appendices.pdf

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Executive Summary

"What gets measured gets managed — even when it's pointless to measure and manage it, and even if it harms the purpose of the organisation to do so"
— V. F. Ridgway, 'Dysfunctional Consequences of Performance Measurements', 1956

"By the time it can be captured in numbers, it's too late."
— Peter Drucker, 'The Effective Executive', 1967

Draft Status: As of **October 18, 2026**, this report has been reviewed by the ARPS Budget & Audit Subcommittee and the Fiscal Sustainability Subcommittee chair, and shared with the chairs of the Leverett School Committee, Finance Committee, and Selectboard. Contact the author: Tim Shores at shorest@arps.org.

How to use this report

The report is organized from broad regional patterns down to district-level details. As a descriptive report, it makes no forecasts and offers no recommendations. One way to use it would be as a companion to discussions that depend on meaningful cost comparisons. You could read straight through to build understanding from high-level trends to selected district details. Or you could use this **quick-start guide** to answer specific questions:

- How does my school district spending compare to Western MA baselines?
- How does Chapter 70 aid compare?
- How does required and actual net school spending compare?
- How have these categories of spending and aid grown over time in comparison with Western MA baselines?

Quick-start guide:

1. Review the cohort comparison tables at the end of the Executive Summary.
2. Review the **Scatterplot of enrollment vs. per-pupil expenditure** in Section 1 to see how all districts are grouped.
3. Locate your district in **Section 3** for detailed comparison of expenses, enrollment, Chapter 70 aid, and net school spending compared to the cohort.

Understanding Per-Pupil Expenditure in Context

Per-pupil expenditure (PPE) provides a flawed lens for understanding school budgets. While this single metric obscures the complexity of school systems — shaped by student needs, facility requirements, governance structures, and historical commitments — it remains an accessible standard for state reporting and public discussion. We might as well try to understand it in full. This report examines PPE patterns across Western Massachusetts districts, organizing them into enrollment-based cohorts to identify trends and provide context for budget deliberations.

The Challenge of Interpretation

In February 2025, an Amherst Town Councilor asked why regional school spending was "increasing faster than the town's revenue, especially given the significant long-term reduction in the number of students we teach."[1] This question reflects a common assumption or hope that school budgets should track with municipal revenue and scale proportionally with enrollment. However, evidence of school budget trends suggests different dynamics at work.

Schools are not like factories, nonprofit organizations or hospitals. Schools are like schools, and we expect to fund and operate them like schools. As a social service, education is fundamentally labor-intensive. National data shows school districts dedicate 80% of budgets to staff and benefits — salaries, health insurance, and retirement contributions — double the 35-40% typical in other organizations.[2] These costs respond to regional and sectoral labor markets rather than local tax revenues, with the cost of education service historically rising faster than general inflation.[3] Fixed and semi-fixed costs — staffing minimums, transportation contracts, facility operations, utilities, debt service, insurance, legal — cannot be reduced proportionally with enrollment. Students eligible for special education have a right to services regardless of overall enrollment, with costs driven by individual needs rather than district size. Charter school transfers result in a net loss of state aid to tuition payments and often concentrate higher-need student populations in the traditional schools.

Approach

This report takes the position that before we set out to understand why we spend so much on schools, we should ask two key questions: What has our budget growth rate been, and what was our starting point? Understanding whether education costs have grown predictably or exceptionally is a question we can answer by looking at data collected by the Commonwealth. This report breaks down PPE and compares expenses by category, state aid, and local contributions across enrollment-based district cohorts. These groupings allow comparison of districts facing similar scale challenges. The goal is to support informed deliberation about education funding within the broader context of regional economic conditions and policy decisions.

Key Finding: Today's high PPE driven by legacy of high PPE and net spending above required spending — not declining enrollment

Analysis of spending patterns in Amherst-Pelham, Amherst, Leverett, Pelham, and Shutesbury reveals an important pattern: per-pupil expenditure in these districts has grown 3-6% from 2009-2024, rates that are comparable to or lower than their Western Massachusetts peers. Current high PPE and net school spending (NSS) is the result of normal growth from a high starting point. In 2009, these districts already maintained PPE significantly above regional averages. With normal growth rates applied to above-average baselines, spending will remain above average. Understanding this shifts the conversation from 'why are costs so high?' to 'how do we manage the school system that we've inherited at 3-6% annual growth?'

Implications

This finding should shift our focus from addressing perceived runaway costs to addressing the change in local fiscal effort to fund our schools. Fiscal effort measures a community's local tax burden for schools relative to its capacity to pay. The Commonwealth calculates this as each district's required local contribution divided by its combined property values and resident income — a community spending 5% of its wealth shows lower effort than one spending 10%, regardless of absolute dollar amounts.

Two districts with identical PPE amounts and growth may differ significantly in their tax bases. This difference will show up in fiscal effort calculations. This is an objective measure of property value and income used to calculate state aid, but effort is also subjective. If fiscal effort becomes too much for local taxpayers, communities face a hard choice. In this scenario, school committees must identify educational needs and purposeful spending within available resources, but their principal concern will remain with the best interests of the district and the students. Leadership on deciding available resources falls to appropriating bodies and, ultimately, to voters.

In the end, we are likely to find that our structural funding challenges will persist until the Commonwealth finds the courage to deliver policy changes that no region, town or school district can achieve alone.

For a good start on how to advocate for meaningful change, see the July 2022 report A Sustainable Future for Rural Schools, the steadfast work of the champions at Rural Schools MA, and the Amherst-Pelham Regional School Committee's Fiscal Sustainability Subcommittee October 2025 report.[4]

References

1. Amherst Indy, "Letter: Correcting Councilor Ryan on the Regional School Budget"
https://www.amherstindy.org/2025/02/07/letter-correcting-councilor-ryan-on-the-regional-school-budget/

2. American Association of School Administrators, "School Budgets 101"
https://www.aasa.org/docs/default-source/resources/reports/school-budgets-101.pdf

3. Where's the Money Gone? Changes in the Level and Composition of Education Spending (1995) and Inflation and the Measurement of School Spending (1996)
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4. Rep. Blais - Rural Schools, Rural Schools Advocacy in Massachusetts, and RSC FSS Report.
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https://go.boarddocs.com/ma/arps/Board.nsf/goto?open&id;=DMQPEL64432C

All data in this report comes from the Massachusetts Department of Elementary and Secondary Education (DESE). See Appendix A for detailed source information (https://github.com/timshores/schools/blob/main/ppe_report/output/fy2025/WMPPE%20Appendices%20%28FY2025%20data%29.pdf).

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Executive Summary (continued)

Total PPE comparison: Western MA enrollment cohorts and selected districts

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above baseline
	Baseline
CAGR = (End/Start)^(1/years) – 1	Below baseline

Table 1

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA (all, excl. Springfield)	$12,210	+4.0%	+4.7%	+6.5%	$22,976
Western MA Tiny (0-100 FTE)	$13,941	+5.3%	+6.1%	+10.5%	$31,804
Western MA Small (101-700 FTE)	$12,418	+4.6%	+5.3%	+6.8%	$25,462
Western MA Medium (701-1400 FTE)	$13,220	+4.1%	+4.8%	+6.5%	$25,059
Western MA Large (1401-10K FTE)	$11,774	+3.9%	+4.6%	+6.4%	$21,765
Outliers (Springfield)	$14,608	+3.7%	+5.6%	+8.7%	$26,053

Table 2

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA Medium (701-1400 FTE)	$13,220	+4.1%	+4.8%	+6.5%	$25,059
Amherst-Pelham Regional	$16,211	+3.5%	+3.5%	+5.1%	$27,899
Amherst	$16,029	+4.0%	+3.8%	+4.7%	$29,984

Table 3

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA Tiny (0-100 FTE)	$13,941	+5.3%	+6.1%	+10.5%	$31,804
Leverett	$15,156	+2.8%	+2.1%	+4.5%	$23,732
Pelham	$14,733	+2.4%	+2.0%	+2.7%	$21,573
Shutesbury	$14,011	+4.3%	+5.8%	+3.4%	$27,389

Table 1 compares PPE and cost growth of Western MA school district cohorts with the benchmark of all Western MA districts. **Tables 2 and 3** benchmark districts against their cohort averages.

Cohorts are organized by enrollment size. Grouping districts by enrollment size enables more meaningful cost comparisons—districts of similar size face similar challenges with administration, staffing, facilities, and program requirements. Detailed information about cohorts follow in **sections 1 and 2**. Detailed comparisons of individual districts to cohorts follow in **sections 2 and 3**.

Why the CAGR threshold is more sensitive

Small differences in growth rates compound over time. A 0.5pp difference may seem modest, but:

- $20,000 PPE growing at 4.0% → $36,019 after 15 years
- $20,000 PPE growing at 4.5% → $38,706 after 15 years
- Gap: $2,687/pupil (7.5% more total growth)

For a 100-student district, that's $268,700 in additional annual spending by year 15. The 0.5pp threshold helps identify districts on meaningfully different long-term trajectories.

Executive Summary (continued)

Chapter 70 Aid (per foundation pupil): Western MA enrollment cohorts and selected districts

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above baseline
	Baseline
CAGR = (End/Start)^(1/years) – 1	Below baseline

Table 4

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA (all, excl. Springfield)	$5,239	+3.6%	+4.7%	+6.2%	$9,292
Western MA Tiny (0-100 FTE)	$4,070	+3.1%	+4.0%	+6.5%	$6,596
Western MA Small (101-700 FTE)	$3,831	+3.0%	+3.7%	+4.5%	$6,135
Western MA Medium (701-1400 FTE)	$5,002	+3.2%	+4.0%	+5.0%	$8,235
Western MA Large (1401-10K FTE)	$5,535	+3.8%	+4.9%	+6.6%	$10,011
Outliers (Springfield)	$9,305	+4.1%	+5.4%	+7.3%	$17,825

Table 5

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA Medium (701-1400 FTE)	$5,002	+3.2%	+4.0%	+5.0%	$8,235
Amherst-Pelham Regional	$5,488	+2.4%	+2.7%	+3.1%	$7,966
Amherst	$4,483	+2.3%	+2.8%	+3.7%	$6,487

Table 6

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA Tiny (0-100 FTE)	$4,070	+3.1%	+4.0%	+6.5%	$6,596
Leverett	$2,303	+3.3%	+3.4%	+8.0%	$3,885
Pelham	$2,743	+2.5%	+2.5%	+7.1%	$4,045
Shutesbury	$3,952	+3.3%	+4.6%	+4.2%	$6,629

Tables 4-6 compare Chapter 70 aid (per foundation pupil) across Western MA school districts organized by enrollment size.

Note about enrollment denominators

Chapter 70 aid and net school spending calculations use foundation enrollment for per-pupil figures, while PPE totals and expense categories use in-district FTE enrollment. These metrics come from different DESE reporting touchpoints:

• **Foundation enrollment** (Chapter 70 profile): All students residing in the district on October 1st of the prior year for whom the district is financially responsible.
• **In-district FTE** (End-of-year-report (EOYR)): Full-year enrollment calculated from total membership days for students either residing in the district, or choiced or tuitioned into the district.

This difference complicates direct comparison between Chapter 70 and expense analyses, but remains valid since each district uses the same denominator type within each metric. The comparison is not apples-to-oranges, but a picture of apples in the morning compared with a picture of apples in the evening. For more information, see the DESE Researcher's Guide: https://www.doe.mass.edu/research/researchers.html

Executive Summary (continued)

Actual NSS above Required NSS (per foundation pupil): Western MA enrollment cohorts and selected districts

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp	Above baseline
	Baseline
CAGR = (End/Start)^(1/years) – 1	Below baseline

Table 7

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA (all, excl. Springfield)	$1,352	+5.6%	+6.8%	+3.3%	$3,225
Western MA Tiny (0-100 FTE)	$4,125	+1.7%	+1.3%	-5.9%	$5,423
Western MA Small (101-700 FTE)	$2,315	+4.5%	+4.5%	-2.0%	$4,646
Western MA Medium (701-1400 FTE)	$2,248	+5.2%	+8.6%	+6.3%	$5,065
Western MA Large (1401-10K FTE)	$900	+6.7%	+6.9%	+3.8%	$2,522
Outliers (Springfield)	-$150	+6.5%	+15.1%	+20.3%	$7

Table 8

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA Medium (701-1400 FTE)	$2,248	+5.2%	+8.6%	+6.3%	$5,065
Amherst-Pelham Regional	$3,407	+4.2%	+4.2%	+4.0%	$6,545
Amherst	$6,169	+4.6%	+5.9%	+6.6%	$12,699

Table 9

Cohort/District	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Western MA Tiny (0-100 FTE)	$4,125	+1.7%	+1.3%	-5.9%	$5,423
Leverett	$7,014	+2.7%	+1.0%	-0.9%	$10,788
Pelham	$7,694	+6.2%	+7.9%	+11.9%	$20,171
Shutesbury	$4,331	+8.2%	+7.8%	+5.7%	$15,220

Tables 7-9 compare Actual NSS above Required NSS (per foundation pupil) - showing local funding effort beyond state requirements.

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Section 1 — Western MA traditional public school district trends

Section 1 examines per-pupil expenditure (PPE) trends across all Western Massachusetts traditional school districts, organized by enrollment-based cohorts.

This section provides a regional overview before diving into cohort-specific details (Section 2) and individual district comparisons (Section 3).

Contents

• **Per-pupil expenditure and growth 2009-2025** compares 2009 to 2025 PPE for Western MA districts.
• **Year-over-Year (YoY) growth rates** shows the rate of PPE change from 2009-2025 for Amherst-Pelham, Amherst, Leverett, Pelham, Shutesbury, and their cohorts.
• **5-year and 16-year CAGR by district and cohort** shows how compound annual growth rate (CAGR) smooths out the volatility of YoY rates of change, making it easier to understand and compare change in PPE over time.
• **Distribution of 2025 enrollment and proposed cohort grouping** shows a histogram of Western MA districts by enrollment with cohorts determined by statistics visualized by the histogram.
• **Scatterplot analysis** shows the relationship between district enrollment and PPE for 2025, with a table of cohort members.
• **Maps** of district locations and 2025 enrollment cohorts, 2025 PPE vs enrollment cohort baseline, and PPE growth (2009-2025) vs enrollment cohort baseline.
• **Statistical associations** between enrollment and PPE.

All districts are grouped into enrollment cohorts based on 2025 in-district FTE using IQR (Interquartile Range) analysis. This grouping enables meaningful peer comparisons while acknowledging that enrollment size is just one factor influencing costs. Districts in the same cohort can differ significantly in wealth, demographics, geography, facility age, and governance—all of which shape costs and outcomes.

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Section 1 — Western MA traditional public school district trends

Per-pupil expenditure and growth: 2009 PPE (lighter) to 2025 PPE (darker segment)

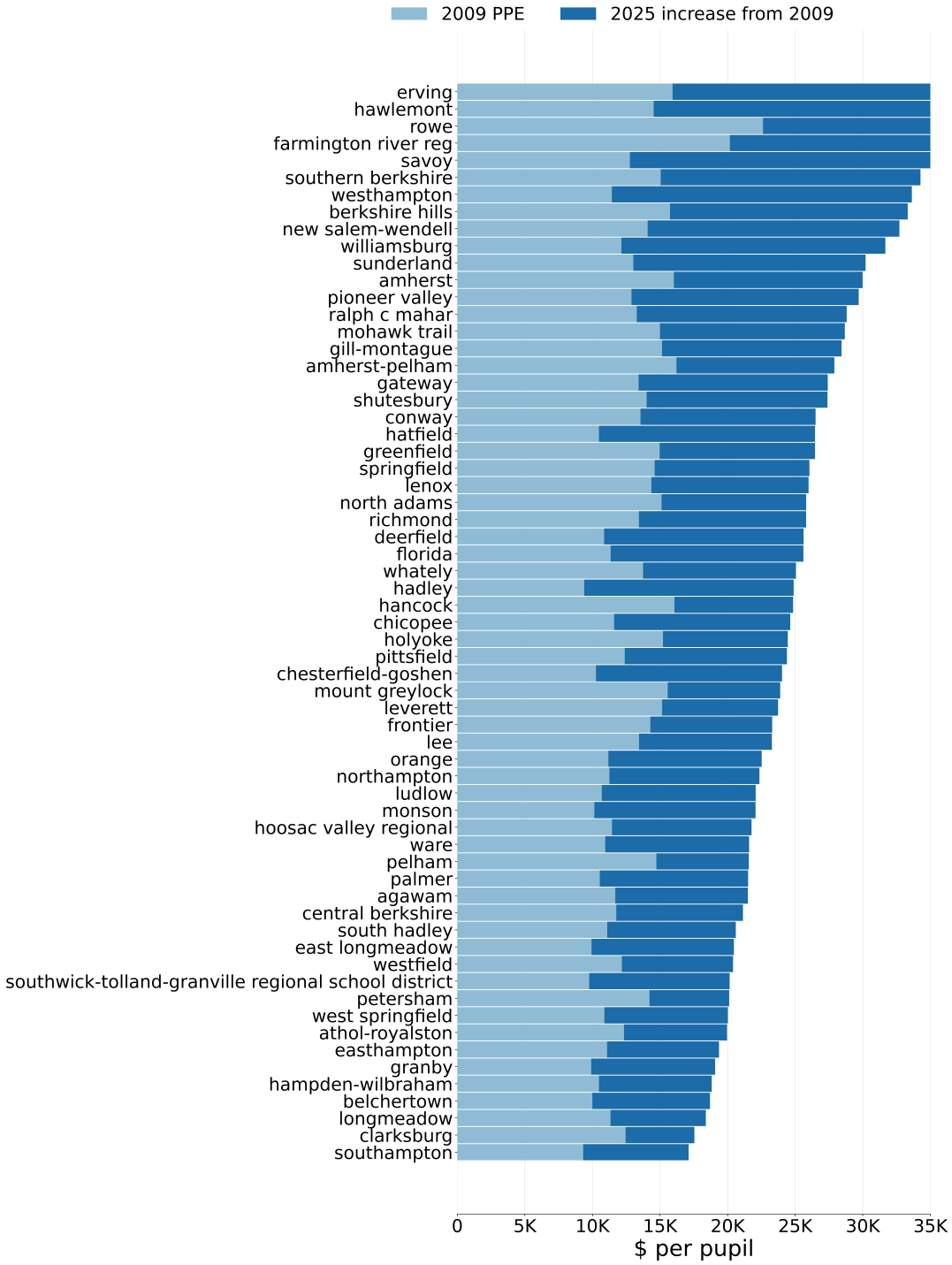


Figure 1

Each bar represents a district's per-pupil expenditures (PPE) in 2025, with growth from 2009 shown by a darker color. Districts are sorted by 2025 spending. While the spread is visible, this alone is nearly meaningless: we don't know to what degree the difference between $15K and $30K per student reflects demographic change, education quality, cost-of-living adjustments, special education demand, economies of scale, administrative decisions, ballot box decisions, or legacy cost structures. The next page zeroes in on rates of change in selected districts.

Note: The following districts are omitted from this analysis: Hampshire (missing expenditure data).

Section 1 — Western MA traditional public school district trends

Year-over-Year (YoY) growth rates by district and cohort

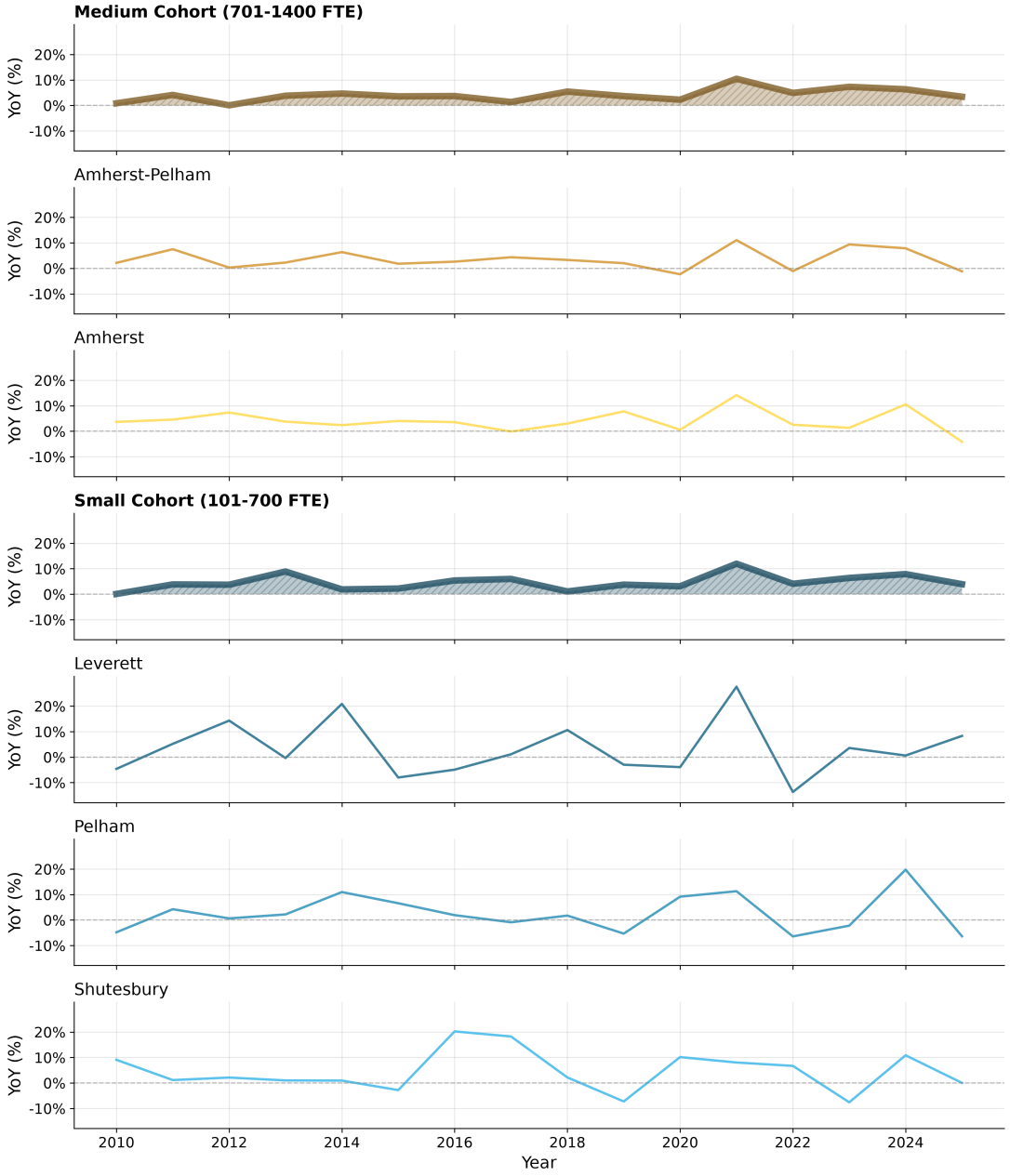


Figure 2

These plots show year-over-year (YoY) growth rates in per-pupil expenditure (PPE) for districts of interest and their enrollment cohorts. Thicker lines represent cohort aggregates, thinner lines show individual districts. Cohort determination is detailed in Appendix A.[1] While patterns emerge, no clear, actionable signal appears across all districts and cohorts over the 2009-2024 period. On the next page, we'll see how a compound growth calculation makes it easier to see similarities and differences between districts.

1. https://github.com/timshores/schools/blob/main/ppe_report/output/fy2025/WMPPE%20Appendices%20%28FY2025%20data%29.pdf

Section 1 — Western MA traditional public school district trends

5-year and 16-year CAGR by district and cohort

Compound annual growth rates (CAGR) smooth out YoY volatility to reveal long-term trends. Below, districts are color-coded by enrollment cohort (blues for Tiny, golds for Medium), with white diagonal lines marking cohort baselines. Comparing three 5-year periods (2010-2015, 2015-2020, 2020-2025) shows how growth rates have shifted over time. The second chart displays overall 16-year CAGR from 2009 to 2025.

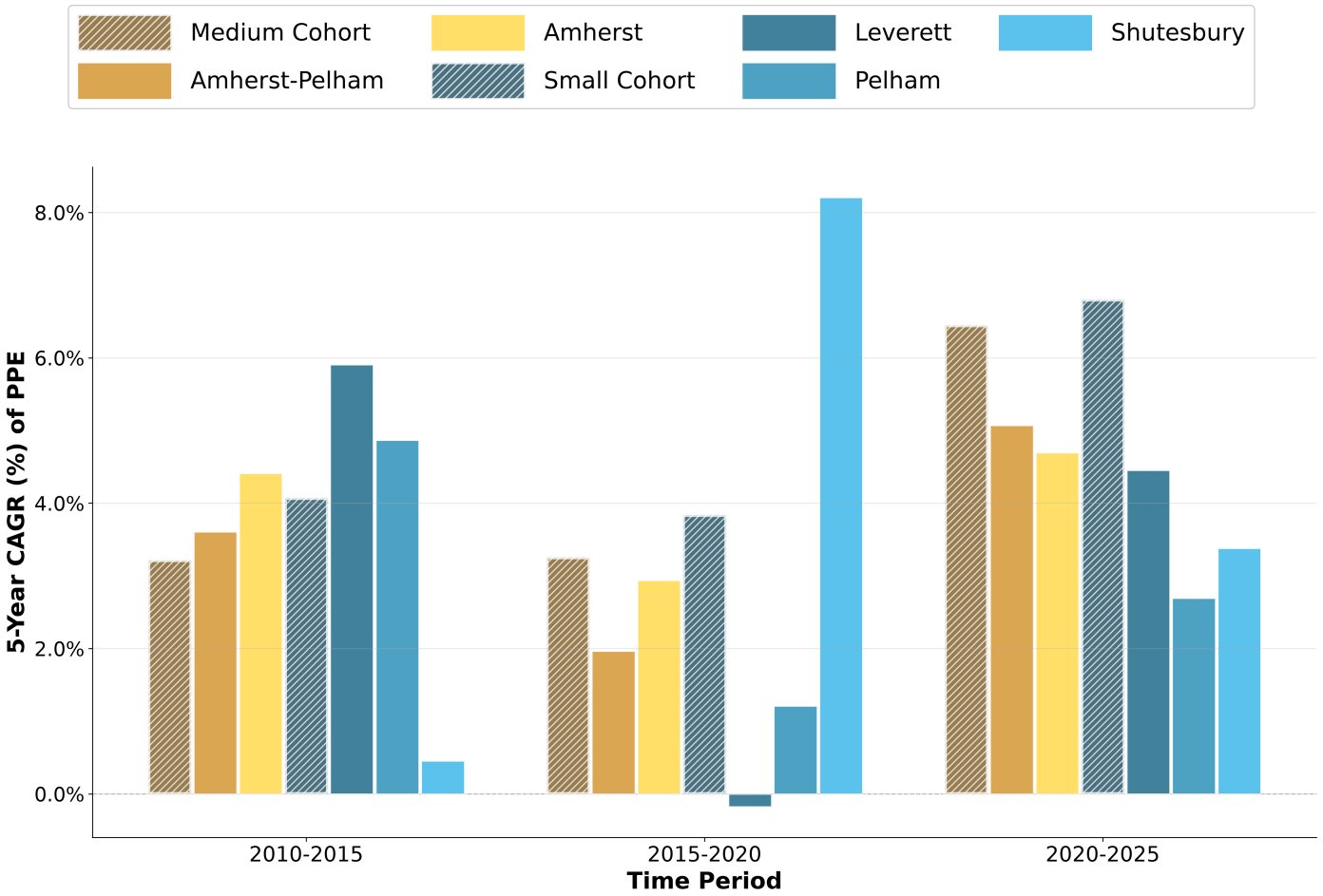


Figure 3

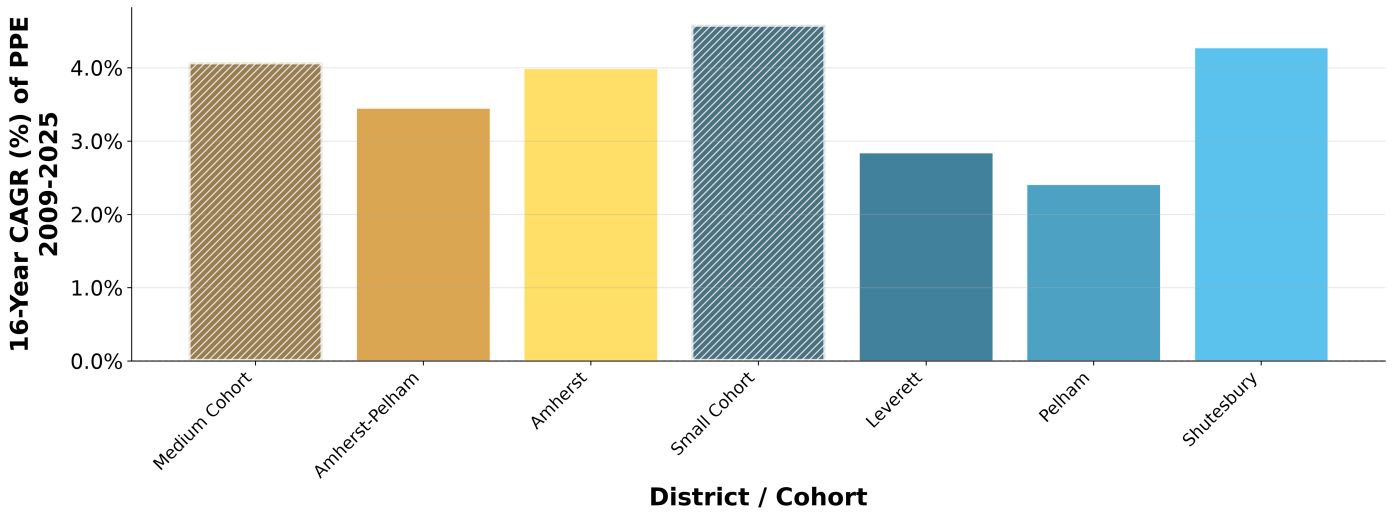


Figure 4

Section 1 — Western MA traditional public school district trends

Distribution of 2025 enrollment and proposed cohort grouping

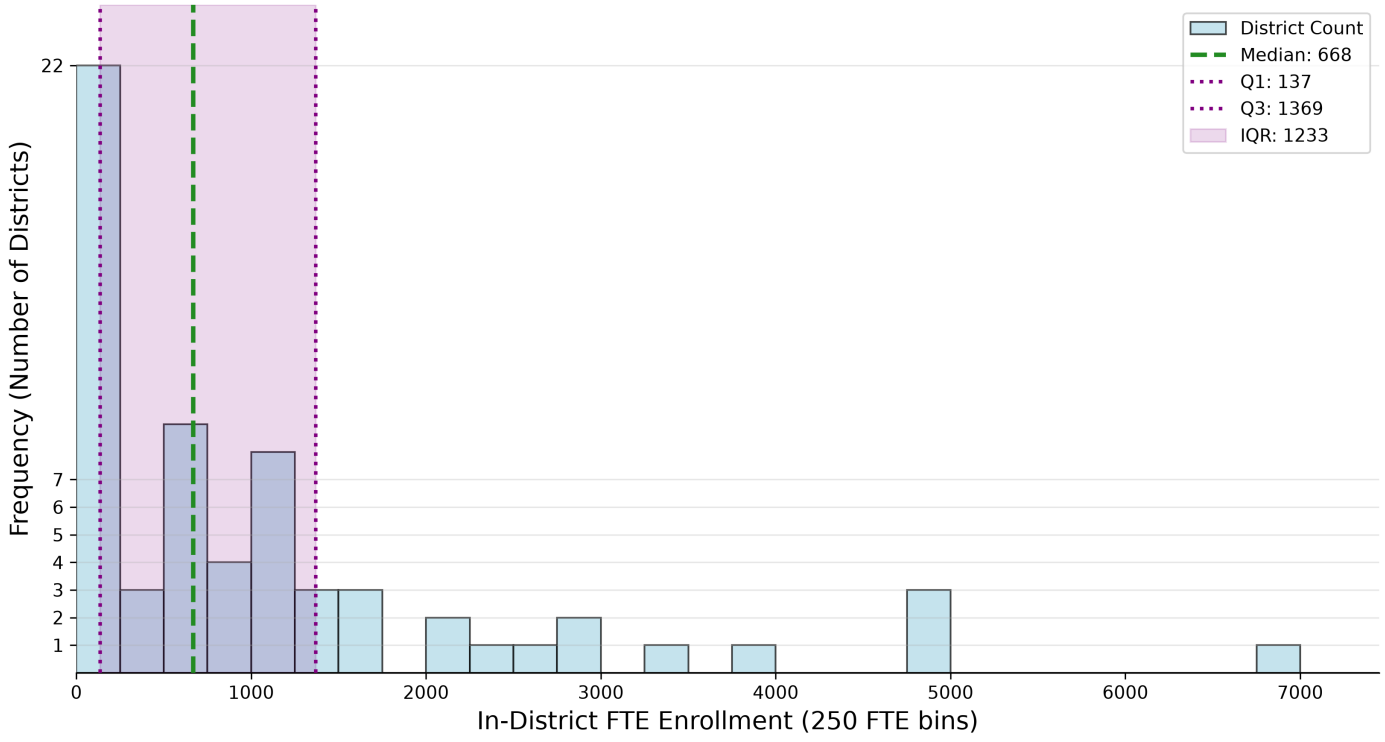


Figure 5

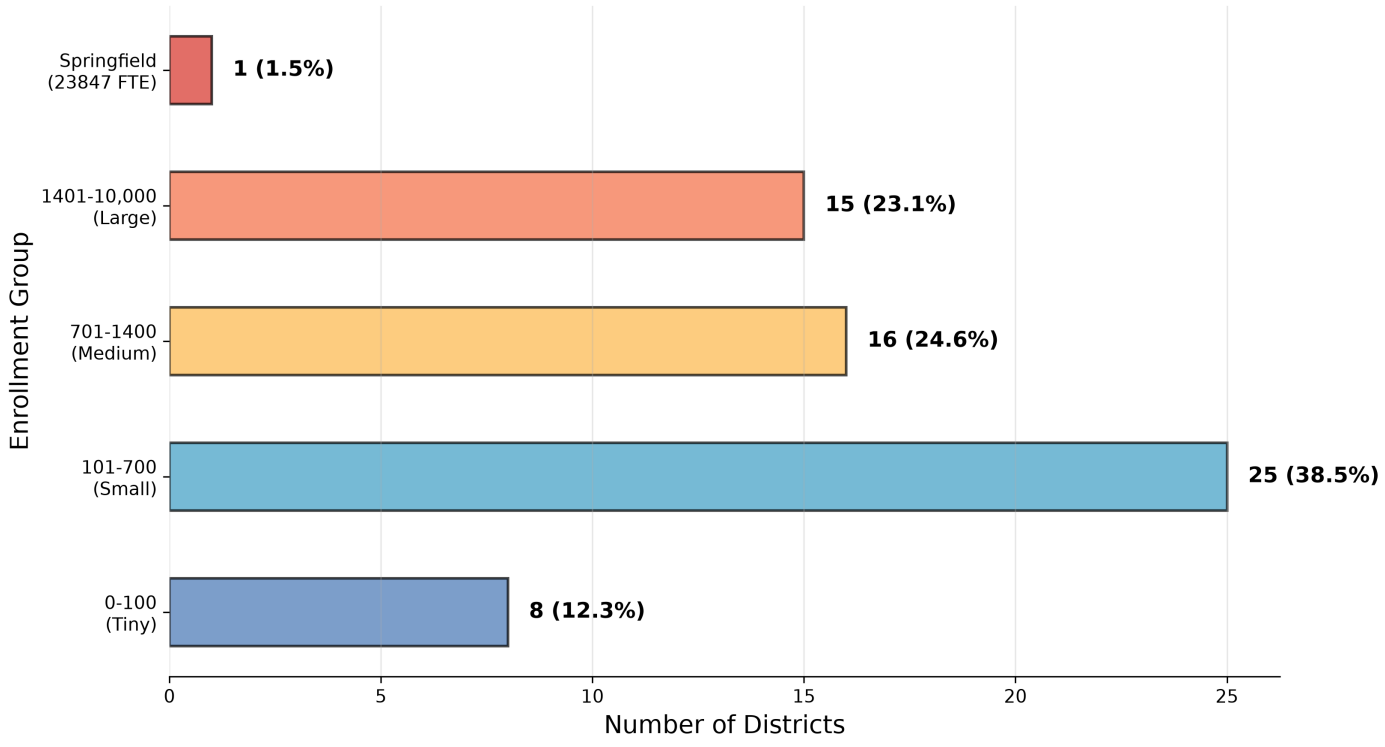


Figure 6

District enrollment cohorts need clear boundaries. To avoid arbitrary cutoffs, this approach uses the natural spread of enrollments. The histogram (top) shows a right-skewed distribution of 2025 enrollments with a long tail of larger districts. Based on interquartile range analysis, districts are grouped into four cohorts (bottom): Tiny (0-200 FTE), Small (201-800 FTE), Medium (801-1600 FTE), and Large (1601-10,000 FTE). Springfield, with an in-district enrollment of 23,723 in 2024, is omitted as an outlier. These cohorts enable comparison by scale to discover financial norms, but are not meant to imply what spending should or should not be for schools of different sizes.

The next page presents each district's enrollment and PPE in both scatterplot and reference table format.

Section 1 — Western MA traditional public school district trends

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2025)

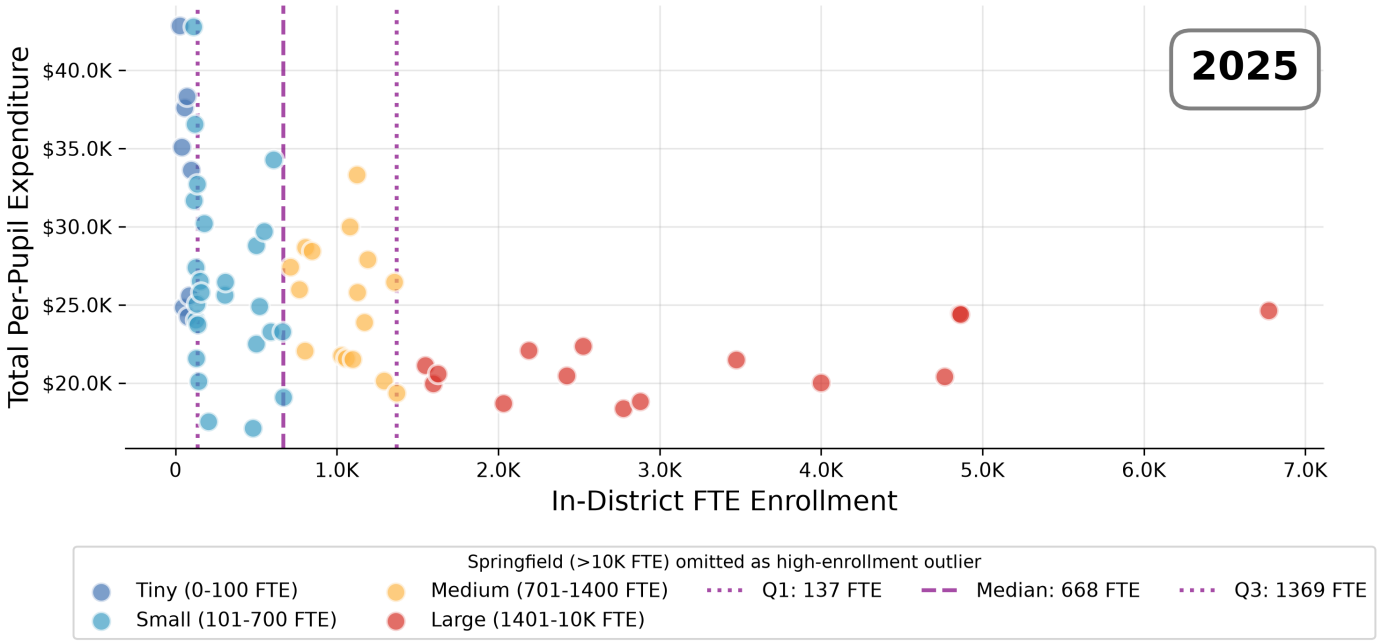


Figure 7

Table 10

District	2025 In-District FTE	2025 PPE ▼
Warwick	28	$42,837
Hawlemont (Regional K-12)	72	$38,330
Rowe	57	$37,601
Savoy	39	$35,071
Westhampton	97	$33,629
Florida	84	$25,605
Hancock	46	$24,840
Worthington	75	$24,251
Erving	112	$42,790
Farmington River Reg (Regional K-12)	121	$36,541
Southern Berkshire (Regional K-12)	607	$34,262
New Salem-Wendell	136	$32,710
Williamsburg	113	$31,667
Sunderland	179	$30,204
Pioneer Valley (Regional K-12)	549	$29,700
Ralph C Mahar (Secondary Region)	501	$28,808
Shutesbury	125	$27,389
Conway	152	$26,509
Hatfield	309	$26,467
Richmond	158	$25,808
Deerfield	304	$25,621
Whately	132	$25,057
Hadley	520	$24,888
Chesterfield-Goshen	127	$24,027
Leverett	137	$23,732
Frontier (Secondary Region)	590	$23,297
Lee	664	$23,272
Orange	500	$22,517
Pelham	129	$21,573
Petersham	142	$20,111
Granby	668	$19,079
Clarksburg	205	$17,544
Southampton	479	$17,111

District	2025 In-District FTE	2025 PPE ▼
Berkshire Hills	1,124	$33,319
Amherst	1,081	$29,984
Mohawk Trail (Regional K-12)	804	$28,665
Gill-Montague (Regional K-12)	845	$28,428
Amherst-Pelham (Secondary Region)	1,189	$27,899
Gateway (Regional K-12)	710	$27,412
Greenfield	1,356	$26,464
Lenox	768	$25,989
North Adams	1,126	$25,809
Mount Greylock (Regional K-12)	1,170	$23,888
Monson	803	$22,065
Hoosac Valley Regional	1,027	$21,768
Ware	1,057	$21,586
Palmer	1,099	$21,514
Southwick-Tolland-Granville Regional School District (Regional K-12)	1,291	$20,152
Easthampton	1,369	$19,360
Chicopee	6,774	$24,634
Holyoke	4,854	$24,453
Pittsfield	4,865	$24,391
Northampton	2,524	$22,356
Ludlow	2,188	$22,079
Agawam	3,473	$21,497
Central Berkshire (Regional K-12)	1,548	$21,135
South Hadley	1,626	$20,603
East Longmeadow	2,423	$20,465
Westfield	4,765	$20,403
West Springfield	4,000	$20,015
Athol-Royalston	1,598	$19,953
Hampden-Wilbraham (Regional K-12)	2,878	$18,820
Belchertown	2,031	$18,699
Longmeadow	2,774	$18,384
Springfield	23,847	$26,053

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Section 1 — Western MA traditional public school district trends

Geographic map showing district locations and enrollment cohorts (2025)

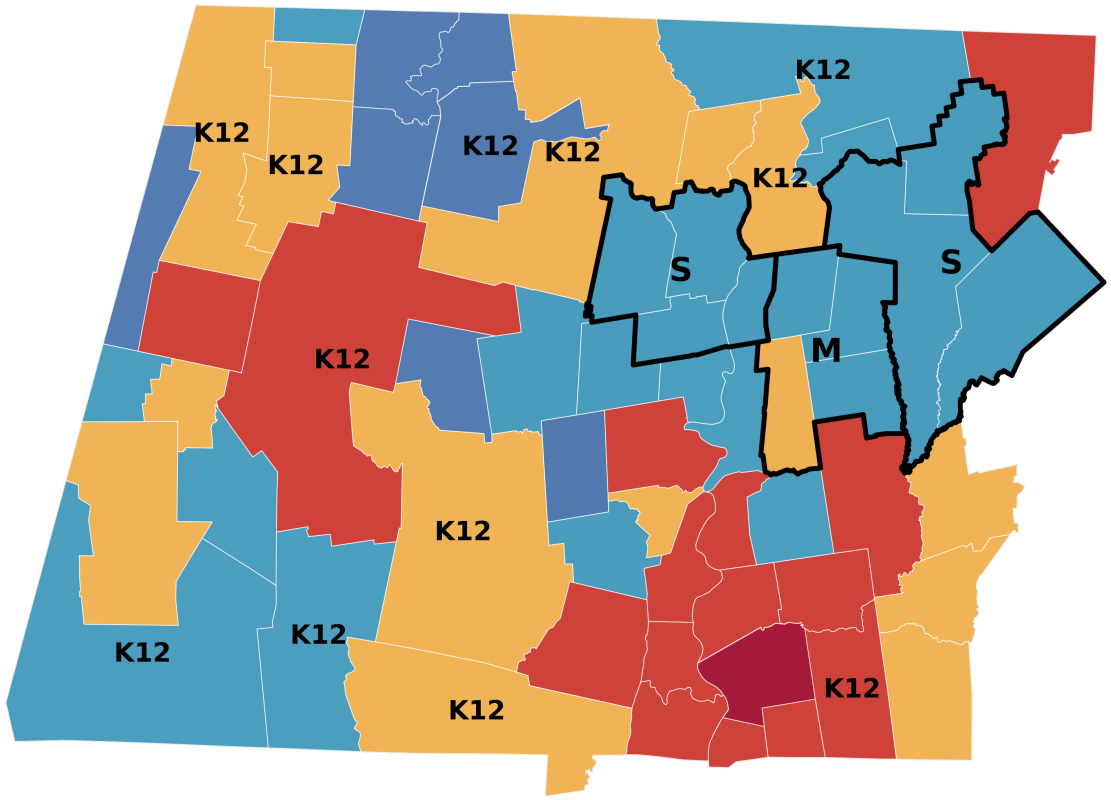


Figure 8

Geography matters for understanding school costs. This map shows where enrollment cohorts cluster and how regional district boundaries cross municipal lines.

Single-town elementary and PK-12 regions appear as solid shaded areas, colored by their enrollment cohort (blue for Tiny, light blue for Small, yellow for Medium, orange for Large). Regional K-12 districts (marked 'K12') span multiple towns. Secondary regional districts are outlined with thick black borders and labeled (T/S/M/L) to show their cohort without covering up their member elementary districts.

See also the MassGIS map of Public School Districts.
https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c

Section 1 — Western MA traditional public school district trends

Geographic map showing 2025 PPE vs enrollment cohort baseline

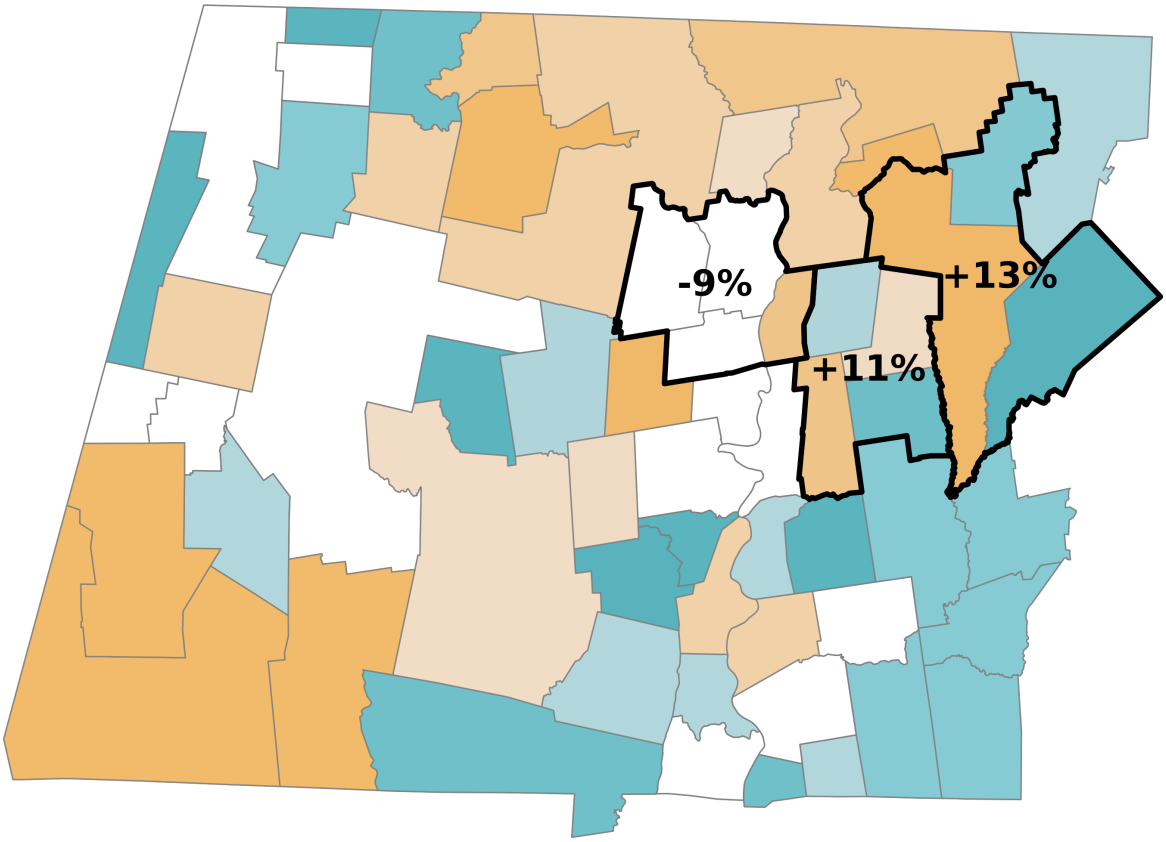


Figure 9

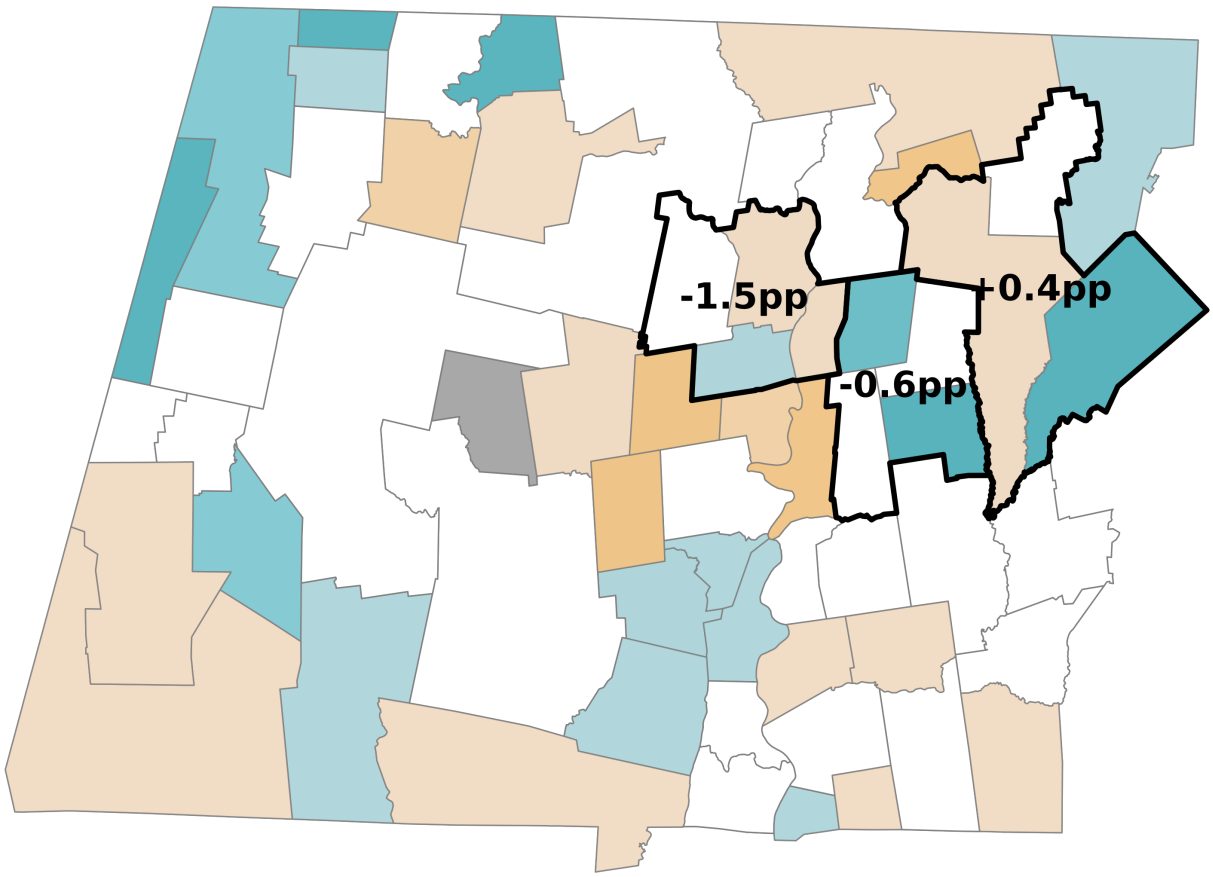
This map shows each district's 2025 per-pupil spending compared to its enrollment cohort baseline (weighted average of all Western MA districts in that cohort). This helps identify which districts are operating above or below the typical spending level for their enrollment size, accounting for the well-documented relationship between district size and per-pupil costs.

Color coding: Dark green indicates districts spending more than 10% below their cohort average (most cost-efficient). Light green shows 0-10% below average. Light red shows 0-10% above average. Dark red indicates districts spending more than 10% above their cohort average (highest relative cost).

See also the MassGIS map of Public School Districts.
https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c

Section 1 — Western MA traditional public school district trends

Geographic map showing 16-year PPE growth (2009-2025) vs enrollment cohort baseline



Total PPE CAGR vs Cohort Baseline - 2009 to 2025 (±0.5pp threshold)

- ≤-2.0pp than cohort: 5 district(s)
- -1.5 to -2.0pp than cohort: 1 district(s)
- -1.0 to -1.5pp than cohort: 3 district(s)
- -0.5 to -1.0pp than cohort: 10 district(s)
- Within ±0.5pp than cohort: 25 district(s)
- +0.5 to +1.0pp than cohort: 13 district(s)
- +1.0 to +1.5pp than cohort: 2 district(s)
- +1.5 to +2.0pp than cohort: 4 district(s)
- Missing data: 1 district(s)

+/-pp = Secondary regional district deviation (n=3)

Figure 10

This map shows each district's 16-year spending growth rate (CAGR, 2009-2025) compared to its enrollment cohort baseline growth rate. This helps identify which districts experienced faster or slower spending growth than similar-sized districts, revealing local policy decisions and cost pressures beyond typical enrollment cohort trends.

Color coding: Dark blue indicates districts with growth more than 0.5 percentage points slower than their cohort average. Light blue shows 0-0.5 pp slower. Light orange shows 0-0.5 pp faster. Dark orange indicates districts with growth more than 0.5 pp faster than their cohort average.

See also the MassGIS map of Public School Districts.
https://massgis.maps.arcgis.com/apps/webappviewer/index.html?id=5b652451857f41ecabc5aa96ef53641c

CLAUDE THIS IS WHERE THE NEW APPENDIX B WILL BEGIN

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Per-Pupil Expenditures

District Size: A Factor, Not the Factor

The five-number summaries and box plots below show two critical dimensions: where districts are now (2025 PPE) and how they got there (2009-2025 PPE growth rates). Box plots show the spread of current spending within each size cohort. Wide boxes indicate greater spending variation; narrow boxes suggest similar spending patterns. The median line shows typical spending for each cohort.

Key finding

Size doesn't tell us much: Enrollment cohort membership explains only 22% of spending variation.

2025 PPE by cohort

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-100 FTE)	8	$24,251	$25,414	$34,350	$37,783	$42,837	
Small (101-700 FTE)	25	$17,111	$23,272	$25,621	$29,700	$42,790	
Medium (701-1400 FTE)	16	$19,360	$21,722	$25,899	$28,031	$33,319	
Large (1401-10K FTE)	15	$18,384	$19,984	$20,603	$22,218	$24,634	
Outliers (Springfield >10K FTE)	1	$26,053	$26,053	$26,053	$26,053	$26,053	

2009-2025 PPE growth rates by cohort

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-100 FTE)	6	2.8%	3.7%	5.7%	6.5%	7.0%	
Small (101-700 FTE)	25	2.2%	3.8%	4.3%	5.4%	6.4%	
Medium (701-1400 FTE)	16	2.7%	3.6%	4.1%	4.6%	5.0%	
Large (1401-10K FTE)	15	3.0%	3.5%	3.9%	4.4%	4.8%	
Outliers (Springfield >10K FTE)	1	3.7%	3.7%	3.7%	3.7%	3.7%	

Similar Growth, Different Starting Points

Growth rates are remarkably consistent across cohorts (3.4-4.6% median CAGR), with the large outlier Springfield growing the slowest. This uniform growth means current spending differences primarily reflect where districts started, not how fast they've grown.

DRAFT

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Chapter 70 Aid (per foundation pupil)

State Aid Doesn't Equalize

This section examines the distribution of Chapter 70 state aid (per foundation pupil) across Western Massachusetts traditional districts organized by enrollment size. The five-number summaries below show two dimensions: current aid levels (2025 Ch70 Aid) and how they grew over time (2009-2025 growth rates).

Key finding

Chapter 70 aid shows a pattern: Medium and Large cohorts receive comparable or higher per-pupil state aid than Tiny districts. Despite this, smaller districts still have higher total PPE — a difference that may be caused by state aid formulas that don't offset enrollment-based cost differences.

DRAFT

2025 Chapter 70 aid per foundation pupil by cohort

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-100 FTE)	8	$3,352	$4,976	$6,503	$8,656	$15,027	
Small (101-700 FTE)	20	$2,829	$4,028	$6,004	$6,700	$12,166	
Medium (701-1400 FTE)	10	$2,895	$6,857	$8,247	$9,513	$11,143	
Large (1401-10K FTE)	11	$2,711	$5,592	$7,483	$12,282	$17,298	
Outliers (Springfield >10K FTE)	1	$17,825	$17,825	$17,825	$17,825	$17,825	

2009-2025 Chapter 70 Aid growth rates per by cohort

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-100 FTE)	7	-5.0%*	2.8%	4.5%	5.7%	6.8%	
Small (101-700 FTE)	20	1.0%	2.7%	2.9%	3.5%	6.2%	
Medium (701-1400 FTE)	10	1.6%	2.4%	3.2%	3.7%	4.0%	
Large (1401-10K FTE)	11	1.2%	2.6%	3.2%	4.1%	4.6%	
Outliers (Springfield >10K FTE)	1	4.1%	4.1%	4.1%	4.1%	4.1%	

* The Worthington district shows anomalous foundation enrollment data, reporting only 8 students in 2009 but 120 students in 2024. Chapter 70 aid increased from $72,731 to $464,966 over this period. However, this represents a decrease in per-pupil aid from approximately $9,091 to $3,875. The unrealistically low foundation enrollment figures prior to 2016 suggest a structural district change or data reporting issue rather than actual enrollment patterns. Such outliers may warrant exclusion from statistical analyses to avoid skewing results.

Section 1 — Western MA traditional public school district trends (continued)

Statistical Associations between Enrollment and Actual NSS (per foundation pupil)

Local Effort Varies Widely

The widest variation appears in discretionary local spending above state requirements. Within the Tiny cohort alone, local effort ranges from $1,275 to $34,082 per pupil above required minimums. This suggests local wealth and priorities — not enrollment — drive spending variation.

2025 Actual NSS above Required NSS per foundation pupil by cohort

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-100 FTE)	8	$-17,100	$4,875	$7,870	$10,765	$30,955	
Small (101-700 FTE)	20	$-16,256	$4,161	$9,861	$14,498	$20,171	
Medium (701-1400 FTE)	10	$512	$2,533	$3,514	$6,754	$22,328	
Large (1401-10K FTE)	11	$-639	$1,314	$2,787	$5,172	$5,898	
Outliers (Springfield >10K FTE)	1	$7	$7	$7	$7	$7	

2009-2025 Actual NSS above Required NSS growth rates by cohort

Cohort	n	Min	Q1	Median	Q3	Max	Distribution
Tiny (0-100 FTE)	7	-10.3%	1.0%	4.5%	6.1%	8.2%	
Small (101-700 FTE)	20	-23.7%	3.1%	6.1%	8.7%	17.0%	
Medium (701-1400 FTE)	10	-0.3%	3.4%	4.4%	5.1%	15.0%	
Large (1401-10K FTE)	11	-12.8%	2.9%	7.7%	10.3%	14.5%	
Outliers (Springfield >10K FTE)	1	-17.4%	-17.4%	-17.4%	-17.4%	-17.4%	

High-spending district patterns

- Historically elevated baselines (high 2009 PPE)
- Moderate growth rates
- Substantial local funding above minimum required spending

Amherst-Pelham exemplifies this: 24% above cohort in 2009, growing slower than peers (3.77% vs 4.09%), yet maintaining position through consistently high local contribution.

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Section 2 — Western MA cohort details

Section 2 provides detailed per-pupil expenditure (PPE) and Net School Spending (NSS) analysis for Western Massachusetts enrollment cohorts: Tiny, Small, Medium, Large, and Springfield. Each cohort section includes:

- Aggregate **expenditure trends by category** (instruction, administration, operations, etc.)
- Aggregate **Chapter 70 aid** and **NSS** patterns over time
- **Enrollment-weighted averages** that represent typical spending across all districts in each cohort

These cohort pages serve as regional baselines for understanding cost patterns by district size.

To skip to **Section 3 — Selected districts**, turn to page 32.

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Section 1: Western MA Traditional District Trends 7
Section 2: Western MA Cohort Details 19
Section 3: Selected Districts 32

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Section 2 — Western MA cohort details

Tiny (0-100 FTE) — PPE and Enrollment — Weighted average per district.

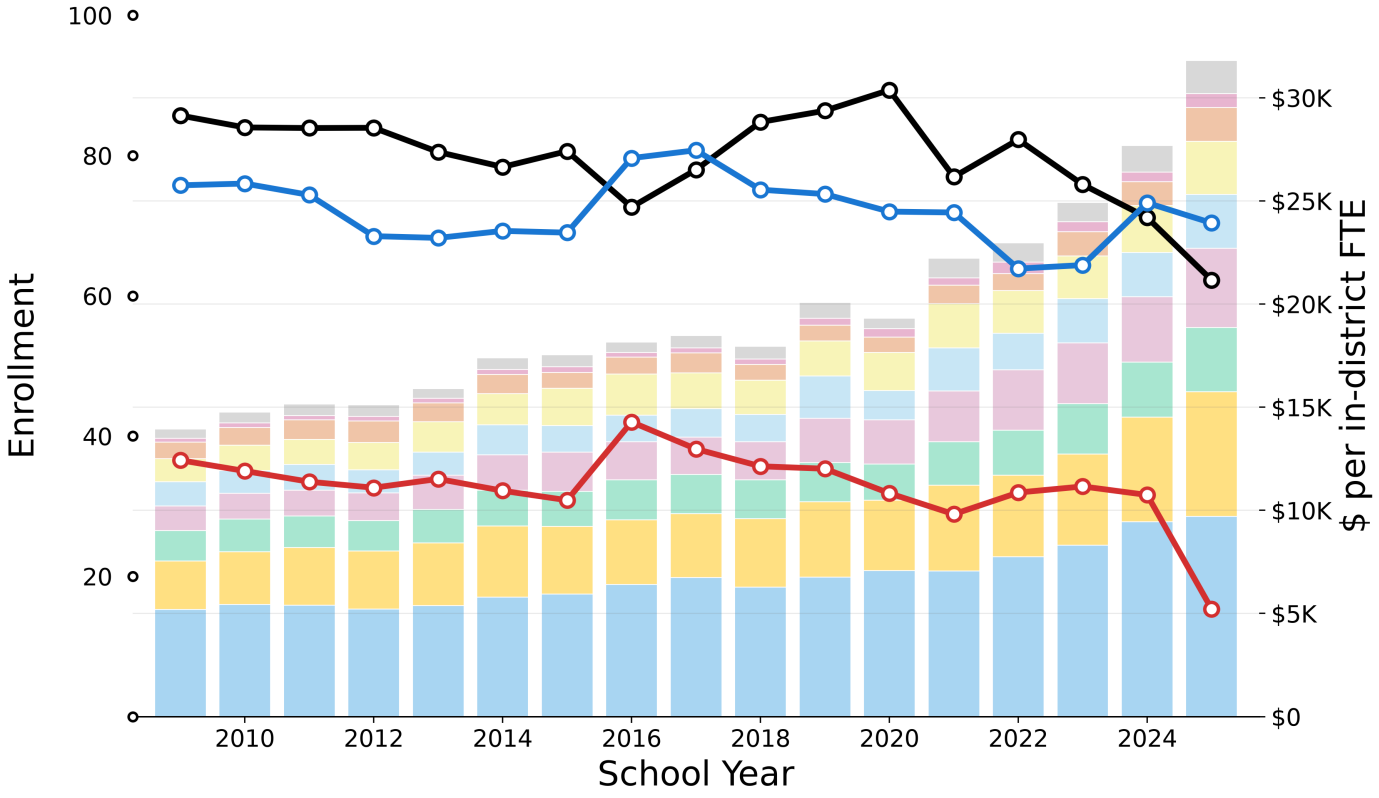


Figure 11

Table 11

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$447	+8.3%	+10.7%	+26.2%	$1,605
Guidance, Counseling and Testing	$195	+8.1%	+9.6%	+10.9%	$674
Administration	$787	+4.7%	+7.8%	+16.8%	$1,637
Instructional Leadership	$1,126	+5.3%	+3.7%	+7.1%	$2,580
Operations and Maintenance	$1,173	+5.1%	+7.2%	+13.0%	$2,610
Other Teaching Services	$1,195	+7.6%	+7.3%	+12.2%	$3,833
Pupil Services	$1,470	+4.8%	+6.2%	+12.1%	$3,123
Insurance, Retirement and Other	$2,362	+6.0%	+6.3%	+12.2%	$6,042
Teachers	$5,186	+4.0%	+5.0%	+6.5%	$9,699
Total	$13,941	+5.3%	+6.1%	+10.5%	$31,804

Table 12

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	86	-2.0%	-2.6%	-7.0%	62
Foundation Enrollment	76	-0.5%	+0.2%	-0.5%	70
Out-of-District FTE Pupils	37	-5.3%	-6.7%	-13.6%	15

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above Western MA baseline

Below Western MA baseline

→ *Compare to PPE and Enrollment for Western MA (all, excl. Springfield) (page 30), Leverett (page 37), Pelham (page 39), Shutesbury (page 41)*

Section 2 — Western MA cohort details

Tiny (0-100 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

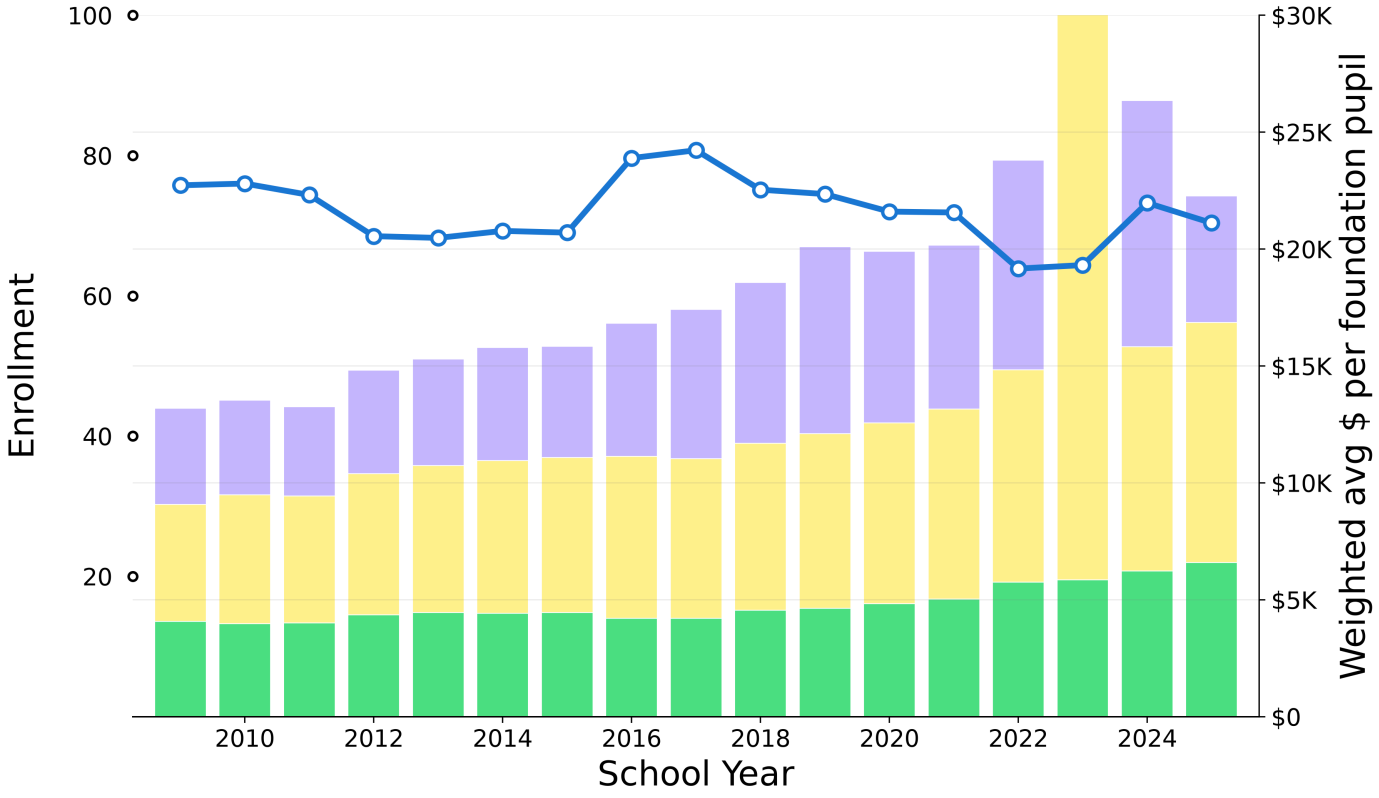


Figure 12

Table 13

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$4,125	1.72%	1.33%	-5.91%	$5,423
Req NSS (minus Ch70)	$5,001	4.59%	4.46%	5.82%	$10,259
Ch70 Aid	$4,070	3.06%	4.00%	6.46%	$6,596
Total (Actual NSS)	$13,196	3.33%	3.47%	2.27%	$22,278
	Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp			Above Western MA baseline	
	CAGR = (End/Start)^(1/years) – 1			Below Western MA baseline	

Tiny (0-100 FTE) includes 8 members:

Districts (6): Florida, Hancock, Rowe, Savoy, Warwick, Worthington.

Regional K-12 (2): Hawlemont, Westhampton.

→ *Compare to Chapter 70 Aid and Net School Spending for Western MA (all, excl. Springfield) (page 31), Leverett (page 38), Pelham (page 40), Shutesbury (page 42)*

Section 2 — Western MA cohort details

Small (101-700 FTE) — PPE and Enrollment — Weighted average per district.

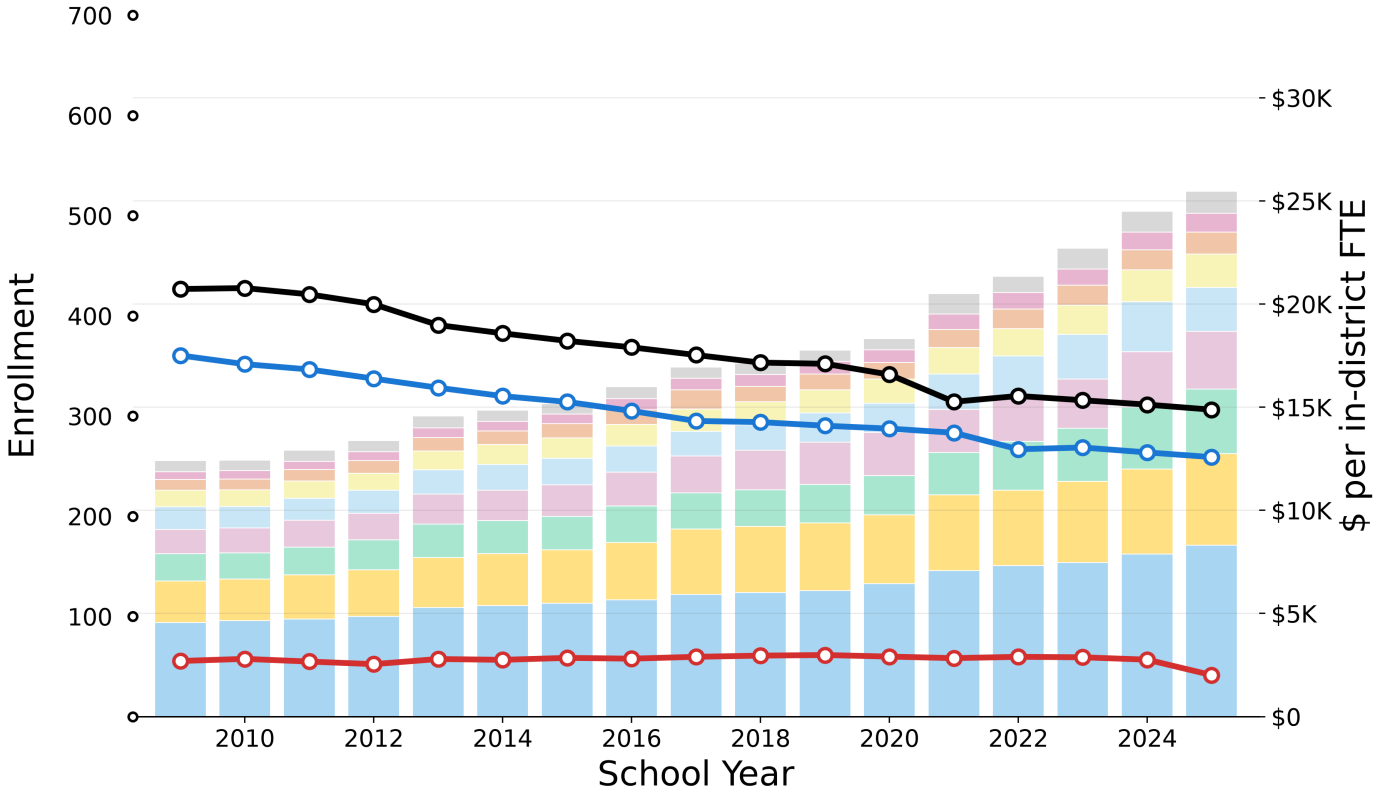


Figure 13

Table 14

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$545	+4.2%	+7.6%	+14.7%	$1,057
Guidance, Counseling and Testing	$377	+5.7%	+6.6%	+8.1%	$910
Administration	$527	+4.6%	+4.7%	+6.1%	$1,077
Instructional Leadership	$800	+4.5%	+5.0%	+6.5%	$1,613
Operations and Maintenance	$1,093	+4.3%	+5.2%	+8.5%	$2,128
Other Teaching Services	$1,185	+5.5%	+6.3%	+6.0%	$2,795
Pupil Services	$1,315	+5.6%	+6.8%	+10.6%	$3,137
Insurance, Retirement and Other	$2,024	+5.0%	+5.5%	+5.8%	$4,444
Teachers	$4,552	+3.8%	+4.2%	+5.2%	$8,302
Total	$12,418	+4.6%	+5.3%	+6.8%	$25,462

Table 15

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	427	-2.1%	-2.0%	-2.1%	306
Foundation Enrollment	360	-2.0%	-1.9%	-2.1%	259
Out-of-District FTE Pupils	55	-1.8%	-3.4%	-7.1%	41

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above Western MA baseline

Below Western MA baseline

Section 2 — Western MA cohort details

Small (101-700 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

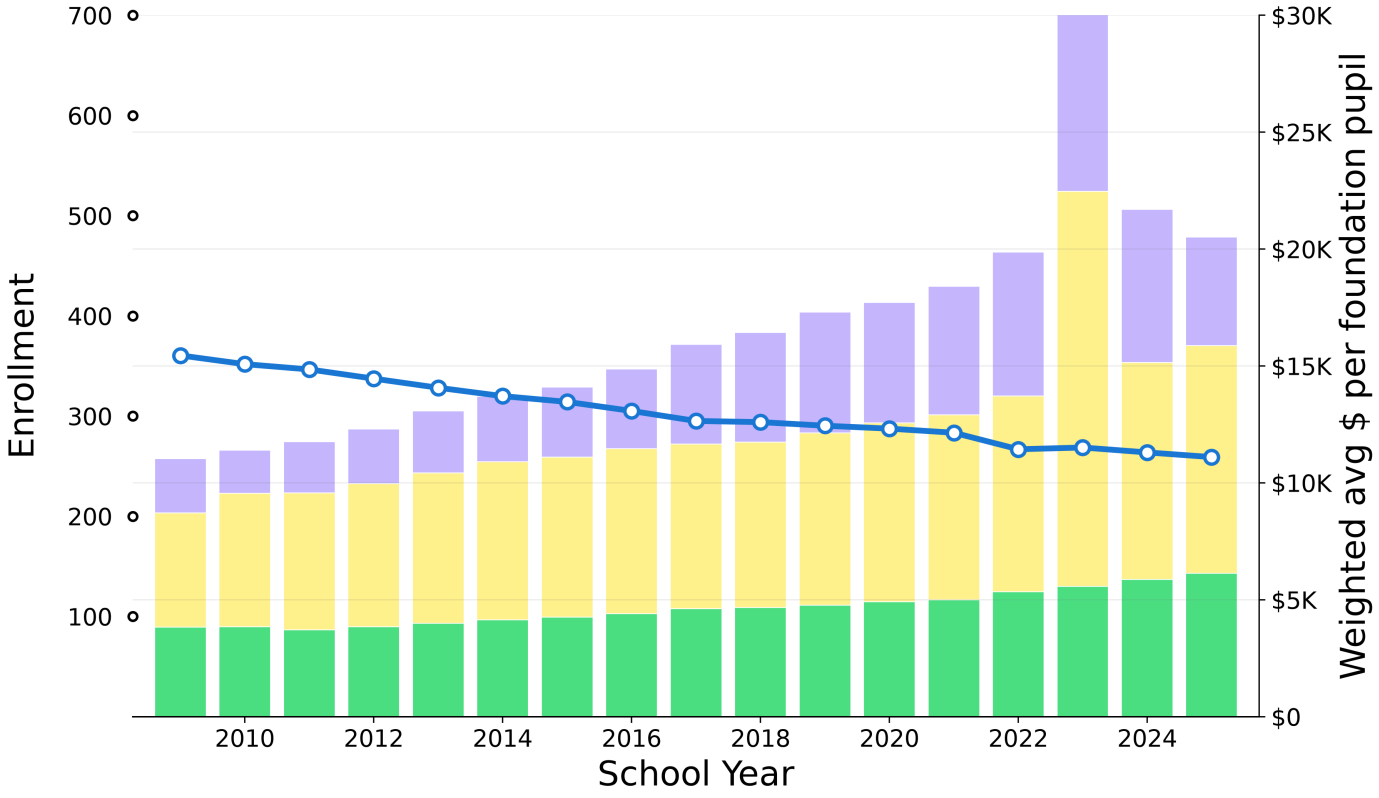


Figure 14

Table 16

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$2,315	4.45%	4.48%	-2.00%	$4,646
Req NSS (minus Ch70)	$4,888	4.40%	3.59%	4.92%	$9,730
Ch70 Aid	$3,831	2.99%	3.74%	4.54%	$6,135
Total (Actual NSS)	$11,033	3.95%	3.83%	2.98%	$20,511

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Small (101-700 FTE) includes 25 members:

Districts (10): Clarksburg, Erving, Granby, Hadley, Hatfield, Lee, Leverett, Pelham, Richmond, Shutesbury.

Regional K-12 (13): Chesterfield-Goshen, Conway, Deerfield, Farmington River Reg, New Salem-Wendell, Orange, Petersham, Pioneer Valley, Southampton, Southern Berkshire, Sunderland, Whately, Williamsburg.

Secondary regions (2): Frontier, Ralph C Mahar.

Section 2 — Western MA cohort details

Medium (701-1400 FTE) — PPE and Enrollment — Weighted average per district.

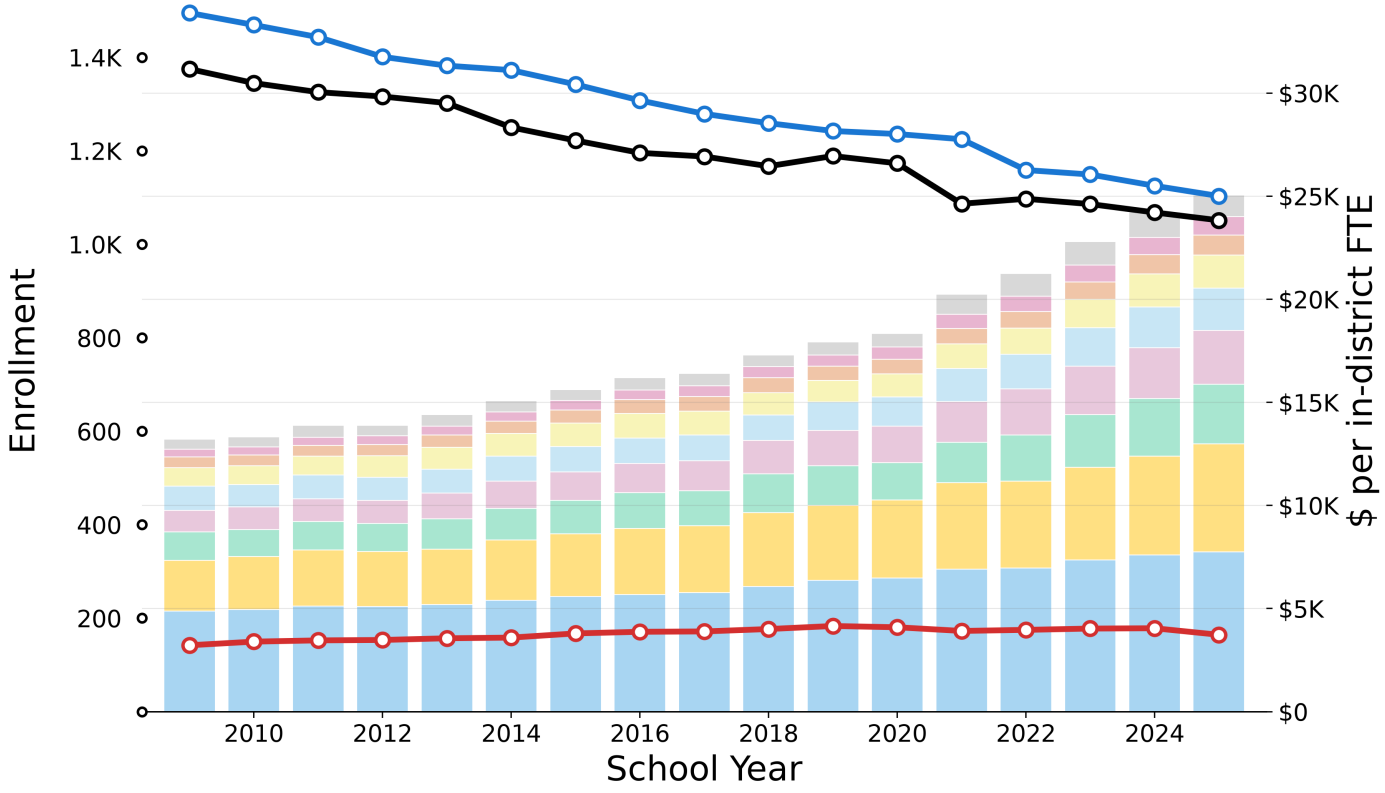


Figure 15

Table 17

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$488	+4.9%	+7.2%	+10.5%	$1,049
Guidance, Counseling and Testing	$382	+5.5%	+6.7%	+8.2%	$897
Administration	$509	+4.1%	+4.4%	+6.4%	$975
Instructional Leadership	$912	+3.6%	+3.6%	+7.5%	$1,601
Operations and Maintenance	$1,172	+3.6%	+5.2%	+7.7%	$2,056
Other Teaching Services	$1,046	+5.8%	+6.6%	+8.1%	$2,594
Pupil Services	$1,363	+4.8%	+6.0%	+9.8%	$2,903
Insurance, Retirement and Other	$2,480	+4.8%	+5.6%	+6.7%	$5,234
Teachers	$4,868	+2.9%	+3.3%	+3.7%	$7,750
Total	$13,220	+4.1%	+4.8%	+6.5%	$25,059

Table 18

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	1,375	-1.7%	-1.5%	-2.2%	1,051
Foundation Enrollment	1,495	-1.9%	-1.9%	-2.2%	1,103
Out-of-District FTE Pupils	142	+0.9%	-0.2%	-1.9%	164

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above Western MA baseline

Below Western MA baseline

→ *Compare to PPE and Enrollment for Western MA (all, excl. Springfield) (page 30), Amherst-Pelham (page 33), Amherst (page 35)*

Section 2 — Western MA cohort details

Medium (701-1400 FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

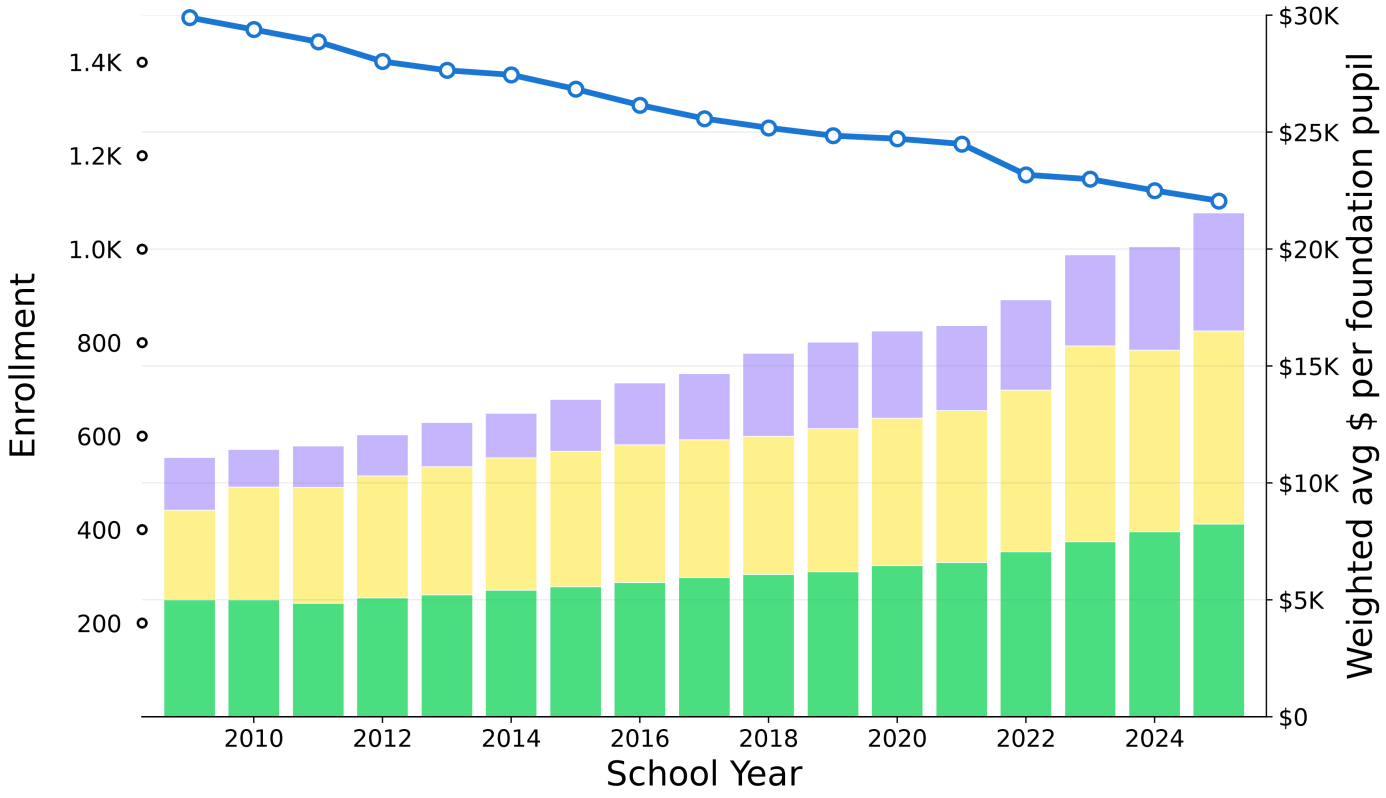


Figure 16

Table 19

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$2,248	5.21%	8.57%	6.30%	$5,065
Req NSS (minus Ch70)	$3,827	4.92%	3.62%	5.55%	$8,252
Ch70 Aid	$5,002	3.17%	4.01%	4.97%	$8,235
Total (Actual NSS)	$11,078	4.25%	4.73%	5.50%	$21,553

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

Medium (701-1400 FTE) includes 16 members:

Districts (9): Amherst, Berkshire Hills, Easthampton, Greenfield, Lenox, Monson, North Adams, Palmer, Ware.

Regional K-12 (6): Gateway, Gill-Montague, Hoosac Valley Regional, Mohawk Trail, Mount Greylock, Southwick-Tolland-Granville Regional School District.

Secondary regions (1): Amherst-Pelham.

→ *Compare to Chapter 70 Aid and Net School Spending for Western MA (all, excl. Springfield) (page 31), Amherst-Pelham (page 34), Amherst (page 36)*

Section 2 — Western MA cohort details

Large (1401-10K FTE) — PPE and Enrollment — Weighted average per district.

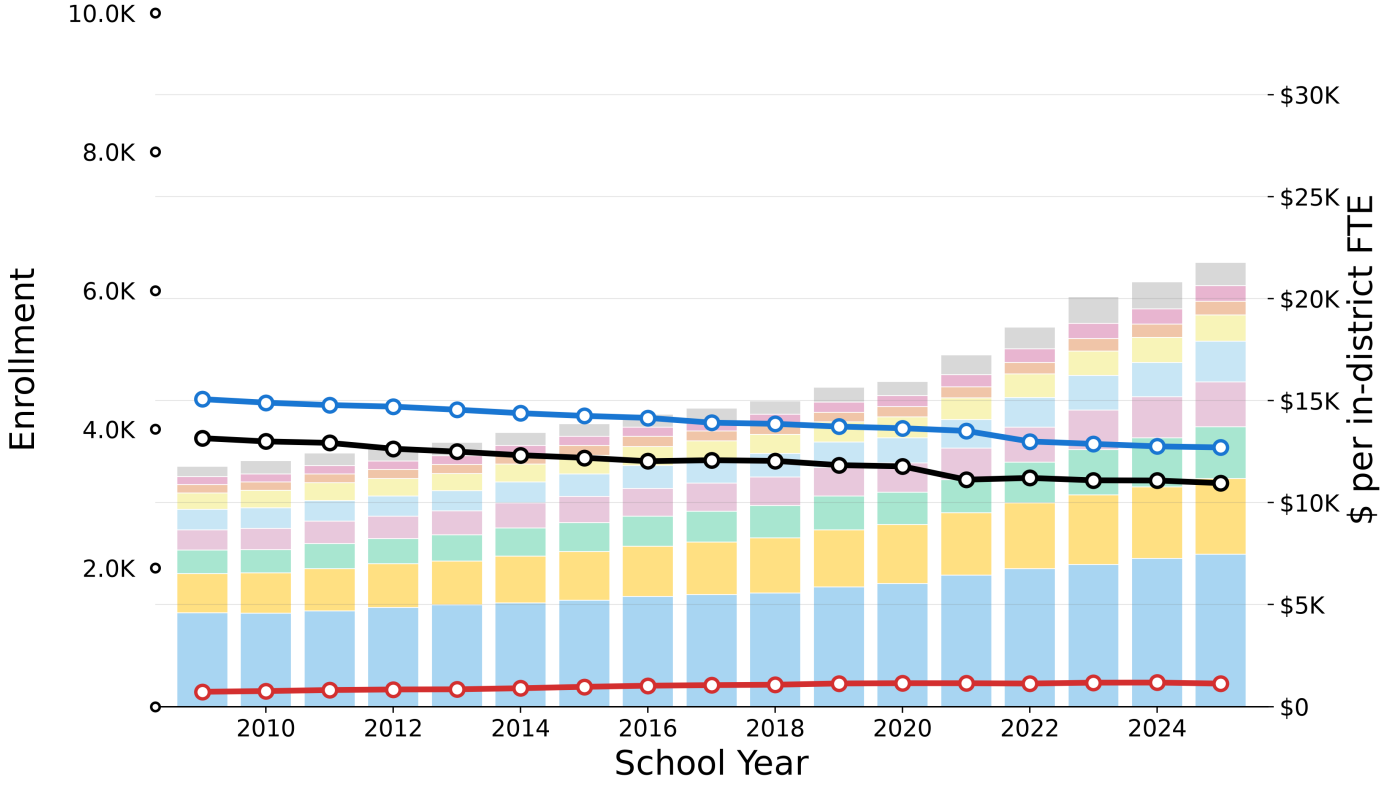


Figure 17

Table 20

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$507	+5.1%	+6.3%	+9.9%	$1,129
Guidance, Counseling and Testing	$393	+4.3%	+5.4%	+7.5%	$773
Administration	$395	+3.4%	+3.3%	+5.9%	$671
Instructional Leadership	$808	+2.9%	+3.4%	+4.5%	$1,281
Operations and Maintenance	$1,017	+4.3%	+6.1%	+10.1%	$1,995
Other Teaching Services	$983	+5.2%	+5.5%	+9.1%	$2,202
Pupil Services	$1,146	+5.1%	+6.0%	+9.7%	$2,534
Insurance, Retirement and Other	$1,927	+4.2%	+4.5%	+5.2%	$3,719
Teachers	$4,598	+3.1%	+3.7%	+4.3%	$7,461
Total	$11,774	+3.9%	+4.6%	+6.4%	$21,765

Table 21

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	3,870	-1.1%	-1.1%	-1.4%	3,221
Foundation Enrollment	4,432	-1.1%	-1.1%	-1.4%	3,736
Out-of-District FTE Pupils	214	+2.8%	+1.5%	-0.3%	331

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) – 1 — Below Western MA baseline

DRAFT

Section 2 — Western MA cohort details

Large (1401-10K FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

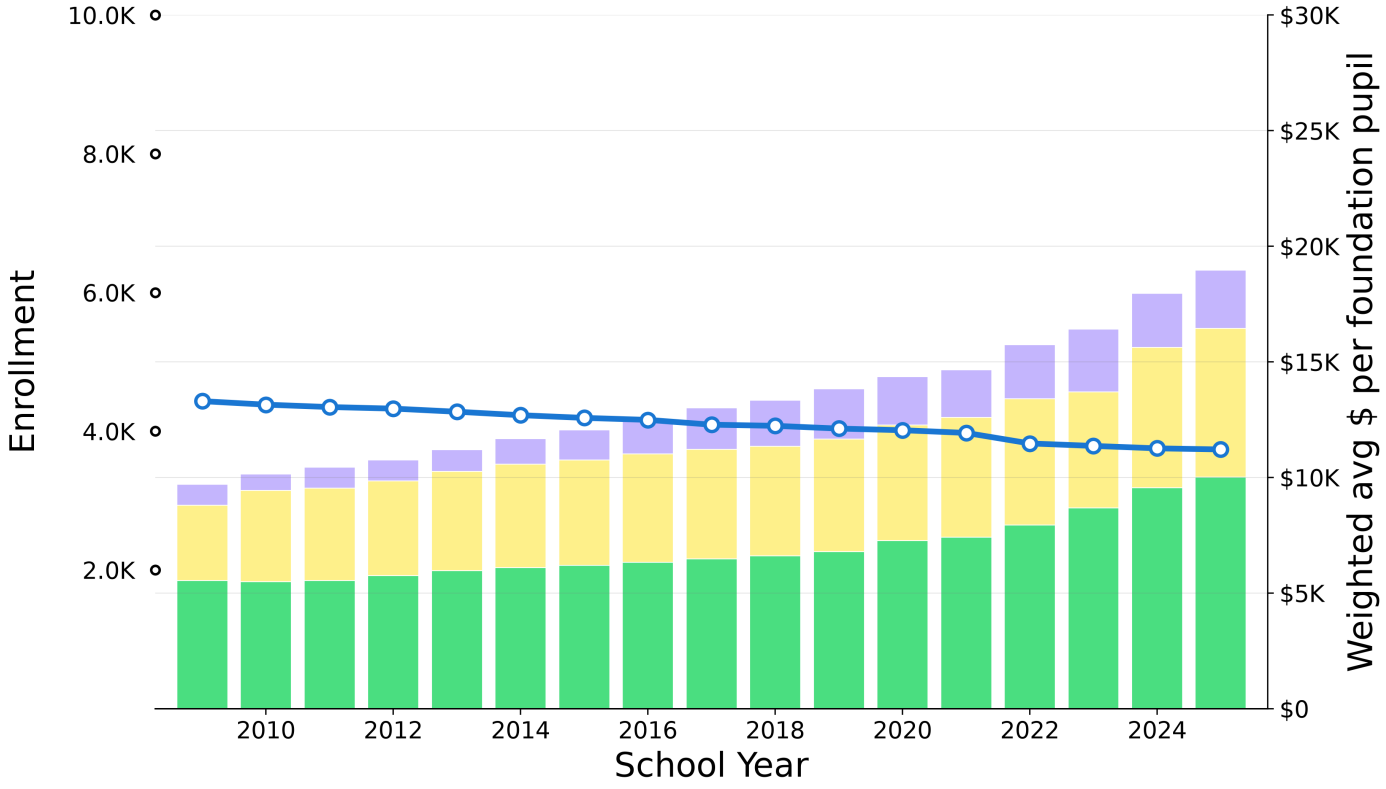


Figure 18

Table 22

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$900	6.65%	6.92%	3.83%	$2,522
Req NSS (minus Ch70)	$3,262	4.34%	3.48%	5.12%	$6,433
Ch70 Aid	$5,535	3.77%	4.93%	6.63%	$10,011
Total (Actual NSS)	$9,697	4.28%	4.64%	5.72%	$18,966

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above Western MA baseline

Below Western MA baseline

Large (1401-10K FTE) includes 15 members:

Districts (12): Agawam, Belchertown, Chicopee, East Longmeadow, Holyoke, Longmeadow, Ludlow, Northampton, Pittsfield, South Hadley, West Springfield, Westfield.

Regional K-12 (3): Athol-Royalston, Central Berkshire, Hampden-Wilbraham.

DRAFT

Section 2 — Western MA cohort details

Outliers (Springfield >10K FTE) — PPE and Enrollment — Weighted average per district.

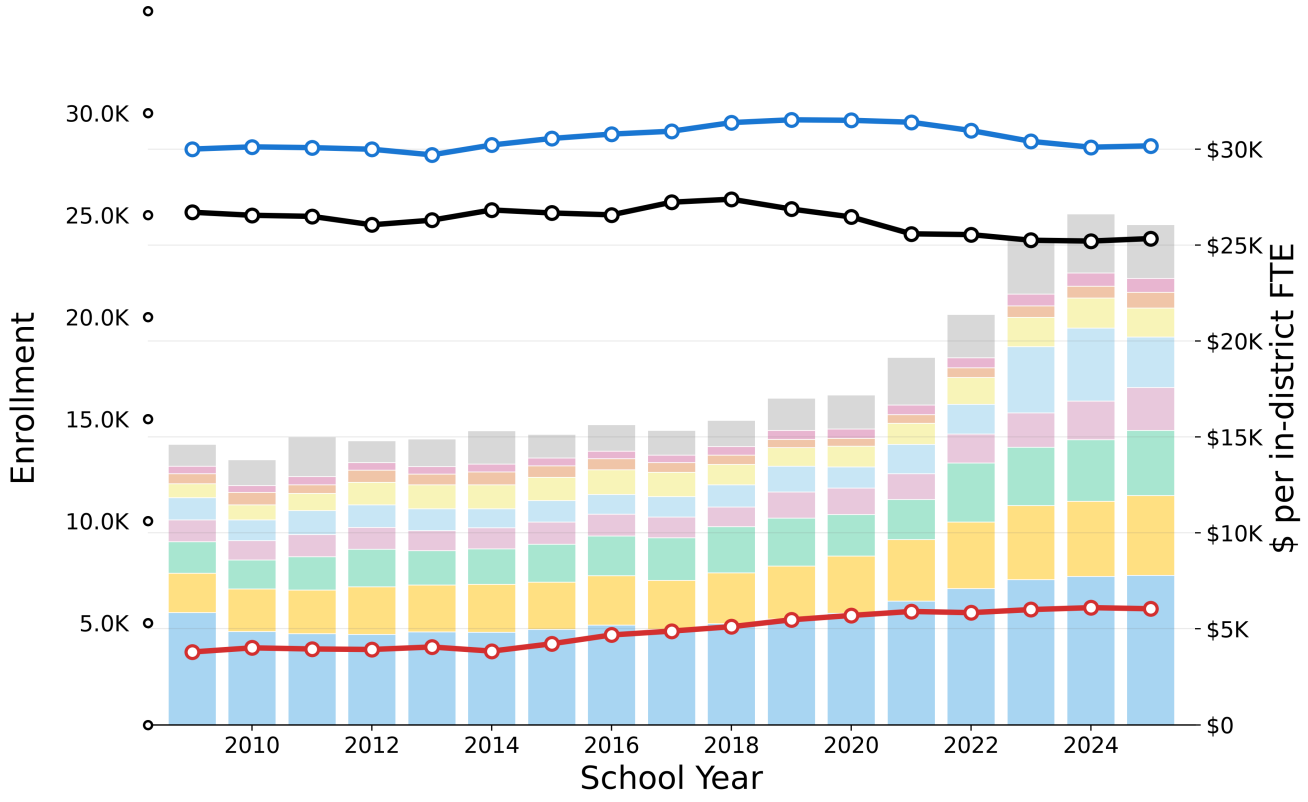


Figure 19

Table 23

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$1,143	+5.7%	+8.4%	+9.6%	$2,792
Guidance, Counseling and Testing	$380	+4.2%	+6.0%	+8.7%	$732
Administration	$533	+2.8%	+3.3%	+15.1%	$823
Instructional Leadership	$719	+4.7%	+2.2%	+6.7%	$1,509
Operations and Maintenance	$1,150	+5.3%	+8.8%	+19.1%	$2,616
Other Teaching Services	$1,143	+4.3%	+6.9%	+10.1%	$2,249
Pupil Services	$1,651	+4.6%	+5.5%	+9.5%	$3,383
Insurance, Retirement and Other	$2,050	+4.5%	+5.4%	+6.8%	$4,165
Teachers	$5,839	+1.8%	+4.6%	+6.1%	$7,784
Total	$14,608	+3.7%	+5.6%	+8.7%	$26,053

Table 24

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	25,137	-0.3%	-0.5%	-0.9%	23,847
Foundation Enrollment	28,235	+0.0%	-0.1%	-0.9%	28,394
Out-of-District FTE Pupils	3,571	+3.0%	+3.7%	+1.2%	5,692

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above Western MA baseline

Below Western MA baseline

DRAFT

Section 2 — Western MA cohort details

Outliers (Springfield >10K FTE) — Chapter 70 Aid and Net School Spending (per foundation pupil).

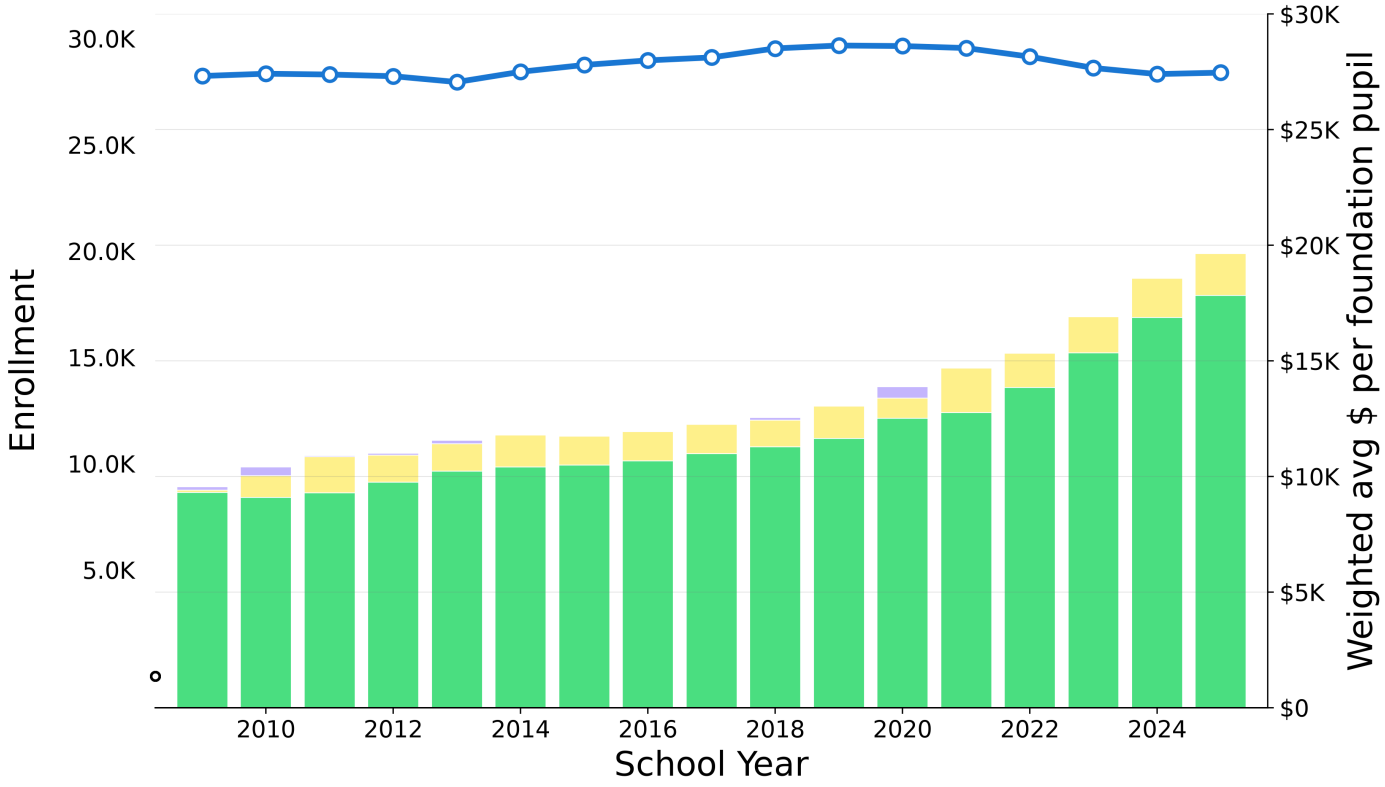


Figure 20

Table 25

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$-150	6.54%	15.05%	20.28%	$7
Req NSS (minus Ch70)	$247	13.26%	3.76%	5.66%	$1,814
Ch70 Aid	$9,305	4.15%	5.45%	7.35%	$17,825
Total (Actual NSS)	$9,403	4.71%	5.28%	7.98%	$19,646

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp — Above Western MA baseline

CAGR = (End/Start)^(1/years) − 1 — Below Western MA baseline

Outliers (Springfield >10K FTE) includes 1 members:

Districts (1): Springfield.

Section 2 — Western MA cohort details

Western MA (all, excl. Springfield) — PPE and Enrollment

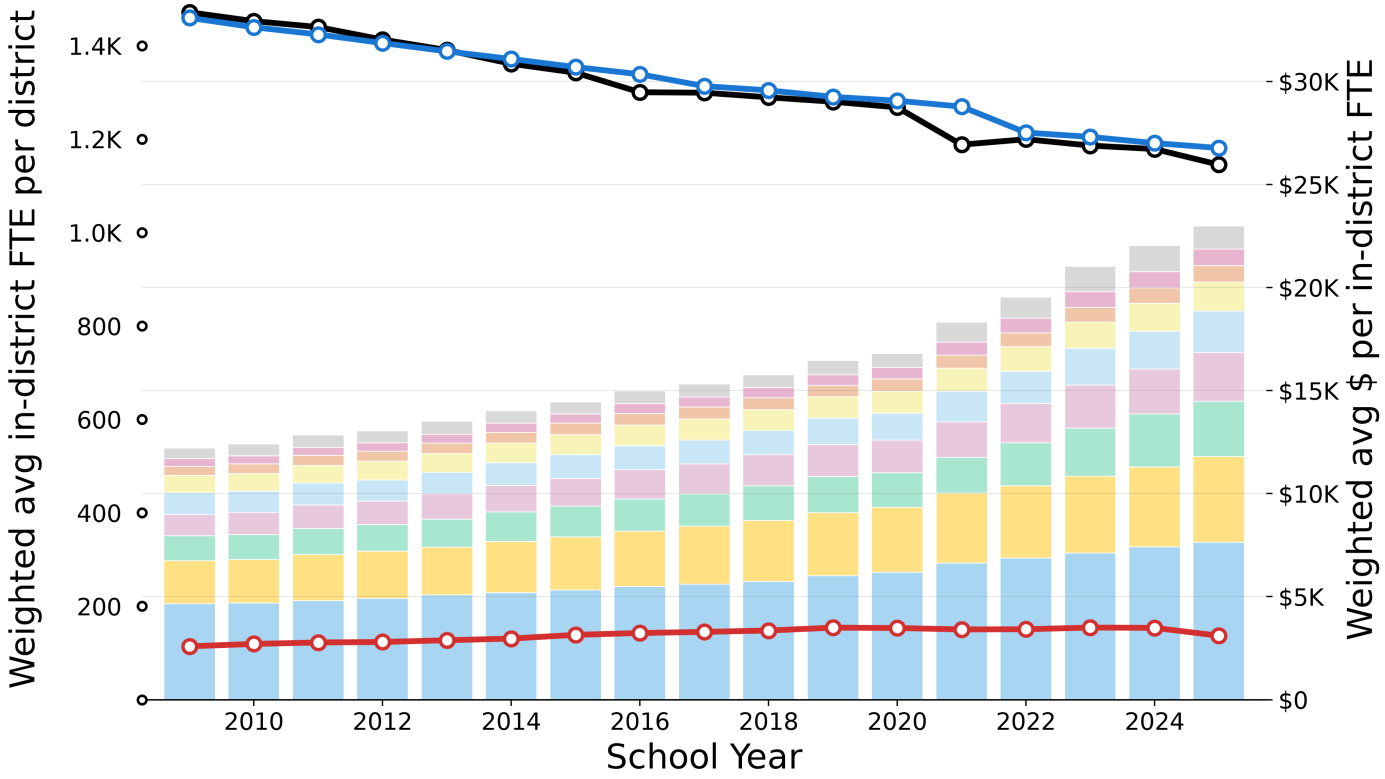


Figure 21

Table 26

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$507	+5.0%	+6.7%	+10.6%	$1,106
Guidance, Counseling and Testing	$387	+4.8%	+5.9%	+7.7%	$815
Administration	$440	+3.7%	+3.8%	+6.1%	$790
Instructional Leadership	$834	+3.3%	+3.6%	+5.5%	$1,398
Operations and Maintenance	$1,064	+4.1%	+5.8%	+9.3%	$2,027
Other Teaching Services	$1,023	+5.4%	+5.9%	+8.4%	$2,365
Pupil Services	$1,220	+5.1%	+6.1%	+9.8%	$2,686
Insurance, Retirement and Other	$2,074	+4.4%	+4.9%	+5.7%	$4,158
Teachers	$4,661	+3.1%	+3.7%	+4.3%	$7,630
Total	$12,210	+4.0%	+4.7%	+6.5%	$22,976

Table 27

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	1,472	-1.6%	-1.6%	-2.0%	1,145
Foundation Enrollment	1,460	-1.3%	-1.4%	-1.6%	1,181
Out-of-District FTE Pupils	114	+1.1%	-0.1%	-2.3%	137

→ *Compare to PPE and Enrollment for Tiny (0-100 FTE) cohort (page 20), Small (101-700 FTE) cohort (page 22), Medium (701-1400 FTE) cohort (page 24), Large (1401-10K FTE) cohort (page 26), Outliers (Springfield >10K FTE) cohort (page 28)*

Section 2 — Western MA cohort details

Western MA (all, excl. Springfield) — Chapter 70 Aid and Net School Spending (per foundation pupil)

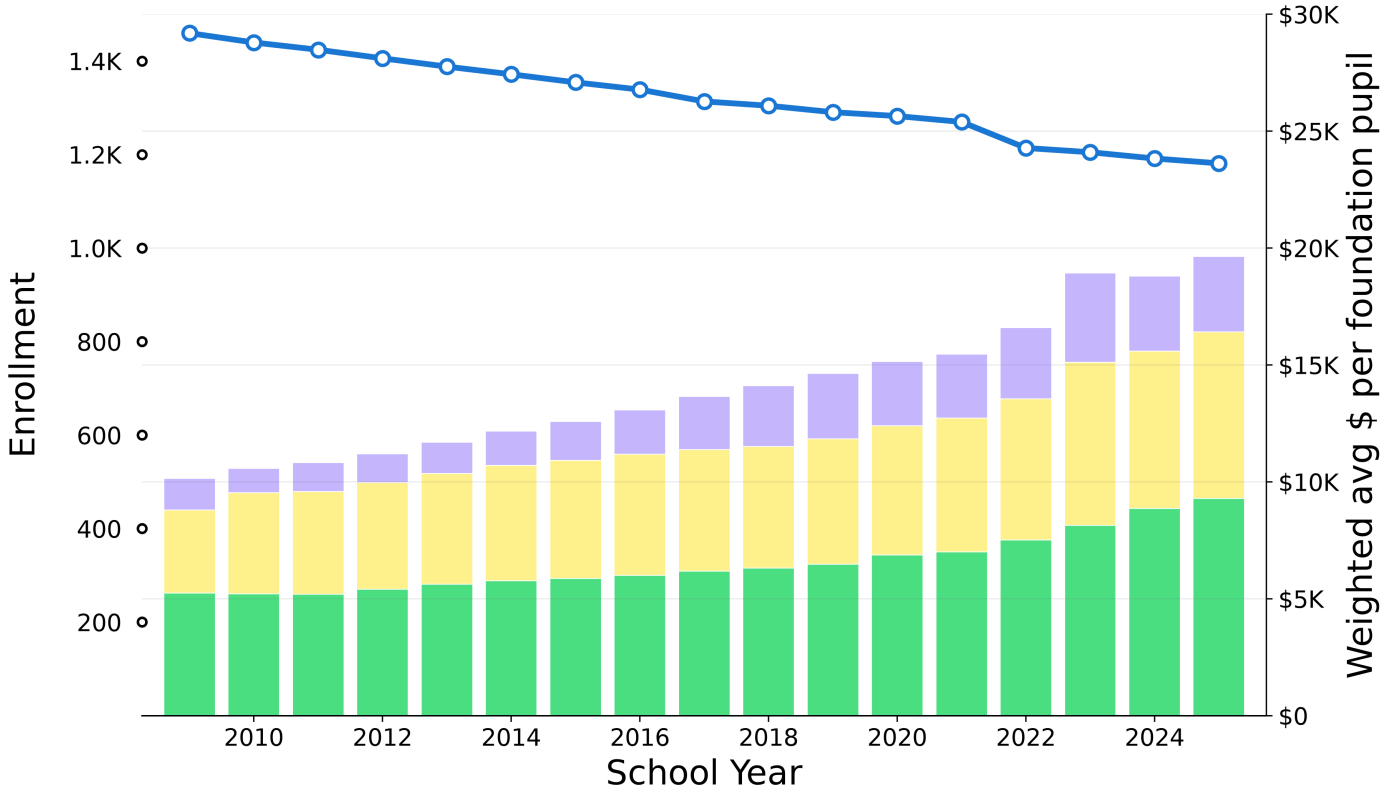


Figure 22

Table 28

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$1,352	5.59%	6.80%	3.31%	$3,225
Req NSS (minus Ch70)	$3,559	4.42%	3.49%	5.16%	$7,112
Ch70 Aid	$5,239	3.65%	4.71%	6.23%	$9,292
Total (Actual NSS)	$10,150	4.21%	4.55%	5.33%	$19,629

Western MA (all, excl. Springfield) includes 64 members:

Districts (37): Agawam, Amherst, Belchertown, Berkshire Hills, Chicopee, Clarksburg, East Longmeadow, Easthampton, Erving, Florida, Granby, Greenfield, Hadley, Hancock, Hatfield, Holyoke, Lee, Lenox, Leverett, Longmeadow, Ludlow, Monson, North Adams, Northampton, Palmer, Pelham, Pittsfield, Richmond, Rowe, Savoy, Shutesbury, South Hadley, Ware, Warwick, West Springfield, Westfield, Worthington.

Regional K-12 (24): Athol-Royalston, Central Berkshire, Chesterfield-Goshen, Conway, Deerfield, Farmington River Reg, Gateway, Gill-Montague, Hampden-Wilbraham, Hawlemont, Hoosac Valley Regional, Mohawk Trail, Mount Greylock, New Salem-Wendell, Orange, Petersham, Pioneer Valley, Southampton, Southern Berkshire, Southwick-Tolland-Granville Regional School District, Sunderland, Westhampton, Whately, Williamsburg.

Secondary regions (3): Amherst-Pelham, Frontier, Ralph C Mahar.

Note: The following districts are omitted from this analysis: Hampshire (missing expenditure data).

→ Compare to Chapter 70 Aid and Net School Spending for Tiny (0-100 FTE) cohort (page 21), Small (101-700 FTE) cohort (page 23), Medium (701-1400 FTE) cohort (page 25), Large (1401-10K FTE) cohort (page 27), Outliers (Springfield >10K FTE) cohort (page 29)

Section 3 — Selected districts

Section 3 examines five selected districts in depth: Amherst-Pelham Regional, Amherst, Leverett, Pelham, and Shutesbury. For each district, three detailed pages show:

• Total PPE breakdown by **expenditure category** with shaded comparison to cohort baseline
• **Chapter 70 aid** and **Net School Spending** breakdown per pupil with shaded comparison to cohort baseline
• **CAGR over 5, 10, and 16-year periods** with shaded comparison to cohort baseline

These district-level pages enable direct comparison between individual district spending patterns and their respective cohorts. Note that although this section focuses on the five selected districts and Western MA, the same visualizations can be produced for other MA districts and regions.

For detailed methodology, see **Appendix A: Data Sources & Calculation Methodology** (https://github.com/timshores/schools/blob/main/ppe_report/output/fy2025/WMPPE%20Appendices%20%28FY2025%20data%29.pdf).

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Section 2: Western MA Cohort Details 19
Section 3: Selected Districts 32

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Section 3 — Specific districts

Amherst-Pelham Regional — PPE and Enrollment

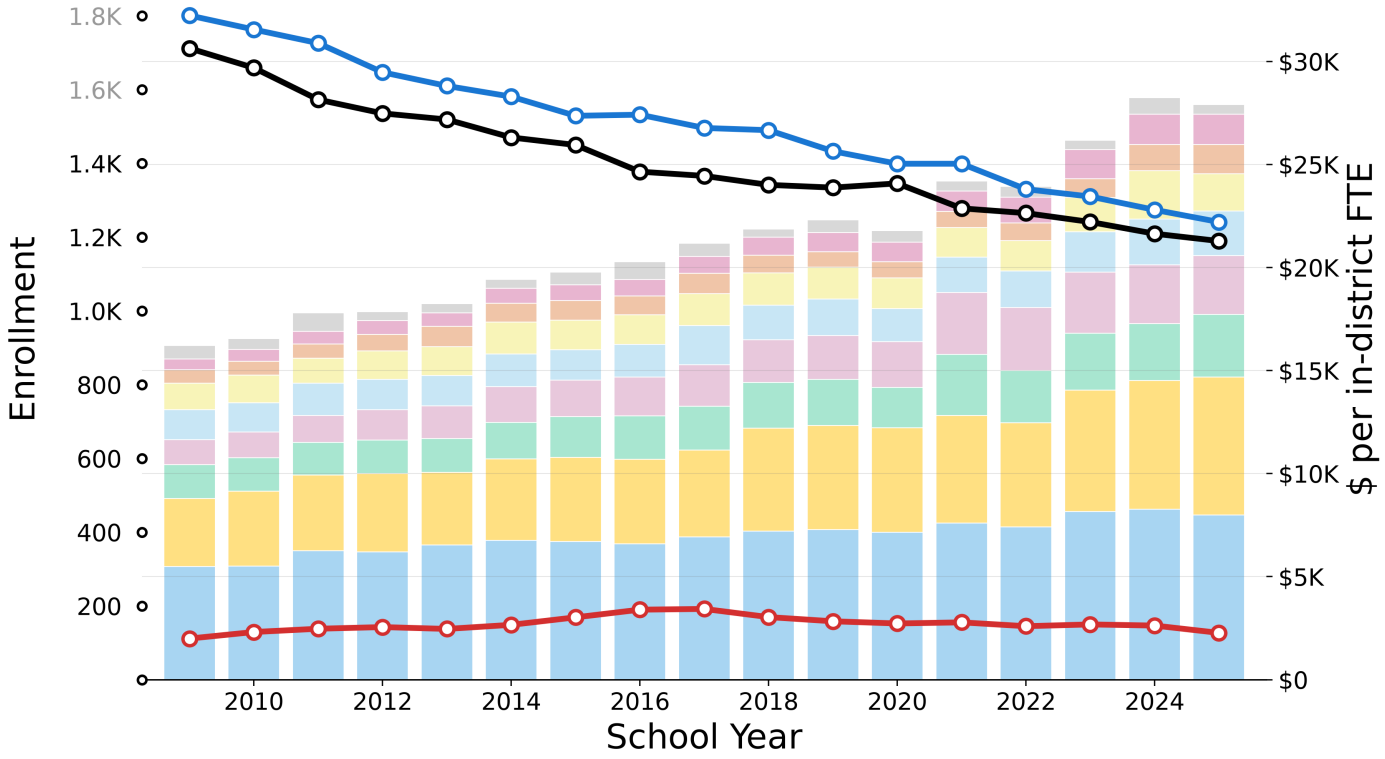


Figure 23

Table 29

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$646	-2.2%	-2.8%	-4.1%	$453
Guidance, Counseling and Testing	$525	+6.7%	+6.6%	+9.3%	$1,471
Administration	$652	+5.1%	+4.2%	+13.0%	$1,435
Instructional Leadership	$1,289	+2.1%	+2.3%	+3.6%	$1,794
Operations and Maintenance	$1,459	+2.5%	+3.9%	+6.3%	$2,168
Other Teaching Services	$1,199	+5.6%	+4.9%	+5.3%	$2,862
Pupil Services	$1,645	+3.9%	+4.4%	+9.2%	$3,039
Insurance, Retirement and Other	$3,305	+4.5%	+5.0%	+5.6%	$6,677
Teachers	$5,491	+2.4%	+1.8%	+2.3%	$8,000
Total	$16,211	+3.5%	+3.5%	+5.1%	$27,899

Table 30

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	1,712	-2.3%	-2.0%	-2.4%	1,189
Foundation Enrollment	1,801	-2.3%	-2.1%	-2.4%	1,241
Out-of-District FTE Pupils	112	+0.8%	-2.8%	-3.6%	127

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above cohort baseline (Medium (701-1400))

Below cohort baseline (Medium (701-1400))

→ *Compare to Tiny (0-100 FTE) cohort: PPE and Enrollment (page 20)*

Section 3 — Selected districts

Amherst-Pelham Regional — Chapter 70 Aid and Net School Spending (per foundation pupil)

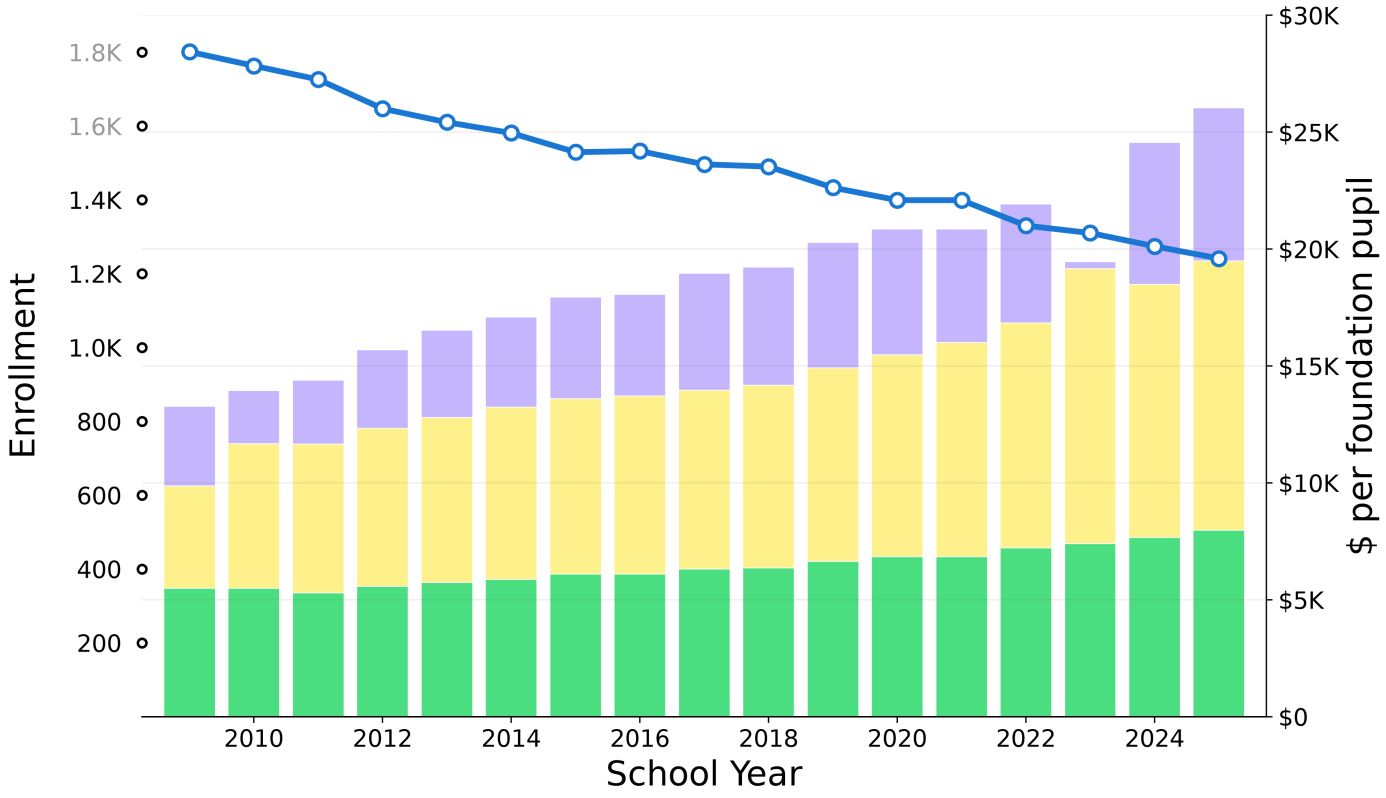


Figure 24 *Table 31*

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$3,407	4.16%	4.18%	3.95%	$6,545
Req NSS (minus Ch70)	$4,383	6.23%	4.38%	5.94%	$11,527
Ch70 Aid	$5,488	2.36%	2.72%	3.11%	$7,966
Total (Actual NSS)	$13,278	4.30%	3.79%	4.53%	$26,038

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) − 1

Above cohort baseline (Medium (701-1400))

Below cohort baseline (Medium (701-1400))

→ *Compare to Tiny (0-100 FTE) cohort: Chapter 70 Aid and Net School Spending (page 21)*

Section 3 — Specific districts

Amherst — PPE and Enrollment

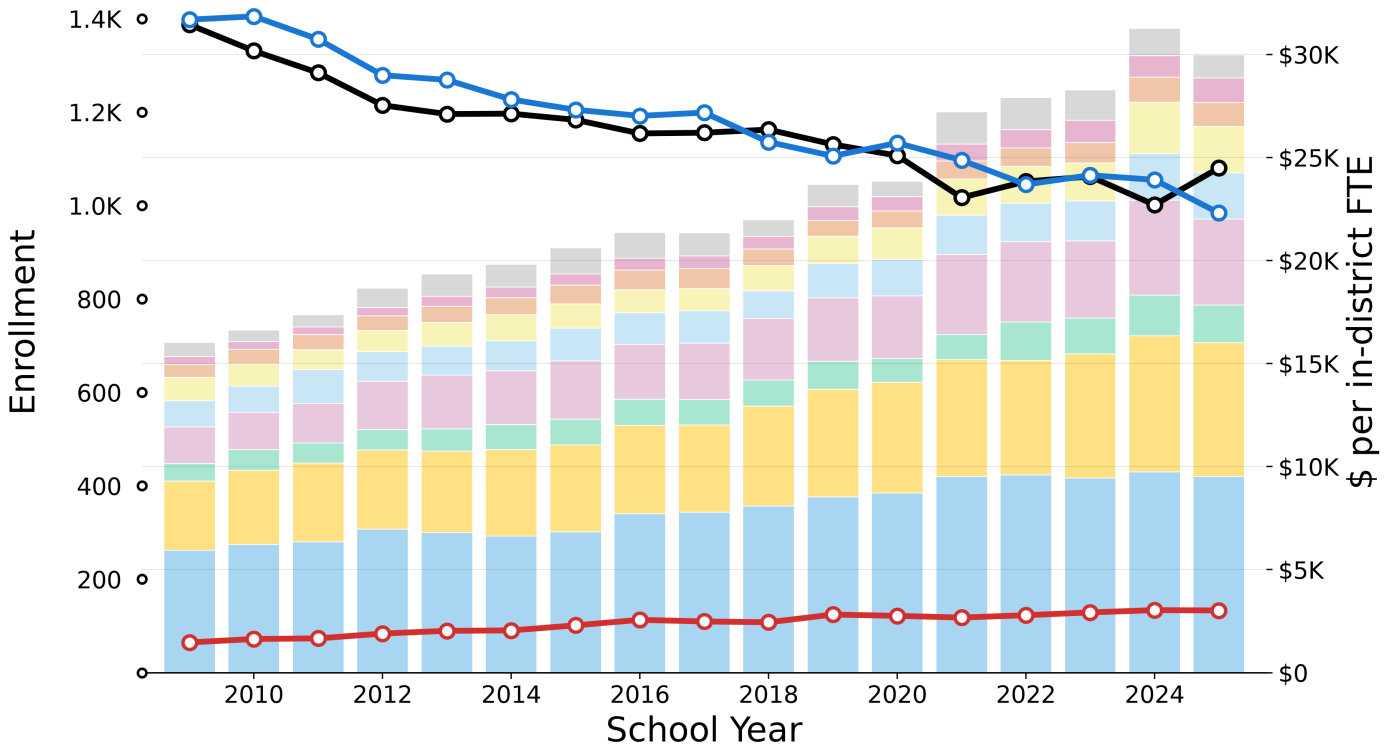


Figure 25

Table 32

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$684	+3.1%	-1.2%	+8.8%	$1,120
Guidance, Counseling and Testing	$396	+7.2%	+8.2%	+11.6%	$1,203
Administration	$627	+3.9%	+2.3%	+7.0%	$1,158
Instructional Leadership	$1,134	+4.4%	+6.7%	+8.1%	$2,253
Operations and Maintenance	$1,264	+3.6%	+3.6%	+4.9%	$2,238
Other Teaching Services	$1,762	+5.5%	+3.9%	+6.5%	$4,176
Pupil Services	$865	+4.8%	+3.8%	+9.7%	$1,827
Insurance, Retirement and Other	$3,353	+4.2%	+4.5%	+3.9%	$6,498
Teachers	$5,944	+3.0%	+3.4%	+1.7%	$9,511
Total	$16,029	+4.0%	+3.8%	+4.7%	$29,984

Table 33

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	1,387	-1.5%	-0.9%	-0.5%	1,081
Foundation Enrollment	1,398	-2.2%	-2.0%	-2.8%	984
Out-of-District FTE Pupils	65	+4.6%	+2.8%	+1.8%	133

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

Above cohort baseline (Medium (701-1400))

CAGR = (End/Start)^(1/years) – 1

Below cohort baseline (Medium (701-1400))

→ *Compare to Medium (701-1400 FTE) cohort: PPE and Enrollment (page 24)*

Section 3 — Selected districts

Amherst — Chapter 70 Aid and Net School Spending (per foundation pupil)

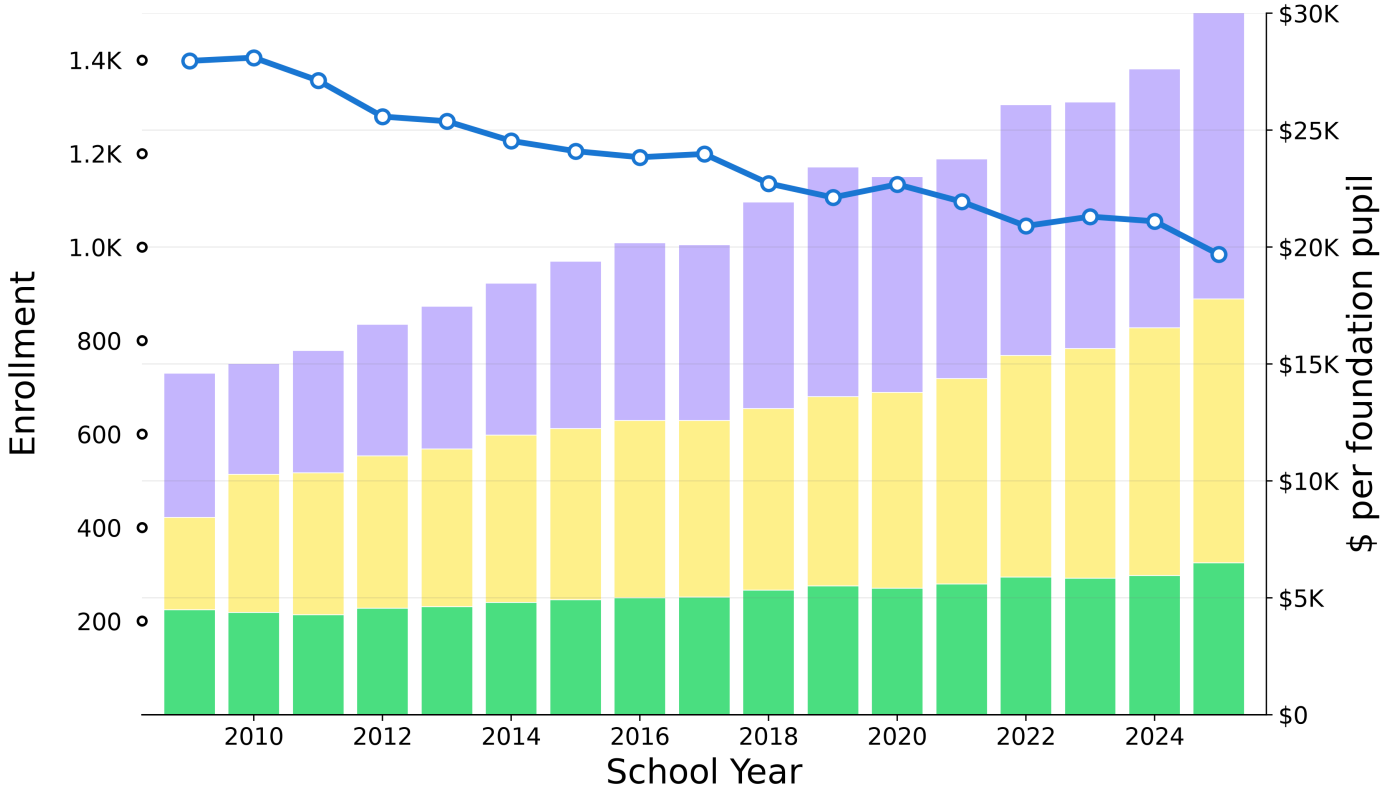


Figure 26

Table 34

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$6,169	4.62%	5.90%	6.59%	$12,699
Req NSS (minus Ch70)	$3,946	6.79%	4.45%	6.15%	$11,299
Ch70 Aid	$4,483	2.34%	2.81%	3.74%	$6,487
Total (Actual NSS)	$14,598	4.71%	4.63%	5.78%	$30,484

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) − 1

Above cohort baseline (Medium (701-1400))

Below cohort baseline (Medium (701-1400))

→ *Compare to Medium (701-1400 FTE) cohort: Chapter 70 Aid and Net School Spending (page 25)*

Section 3 — Specific districts

Leverett — PPE and Enrollment

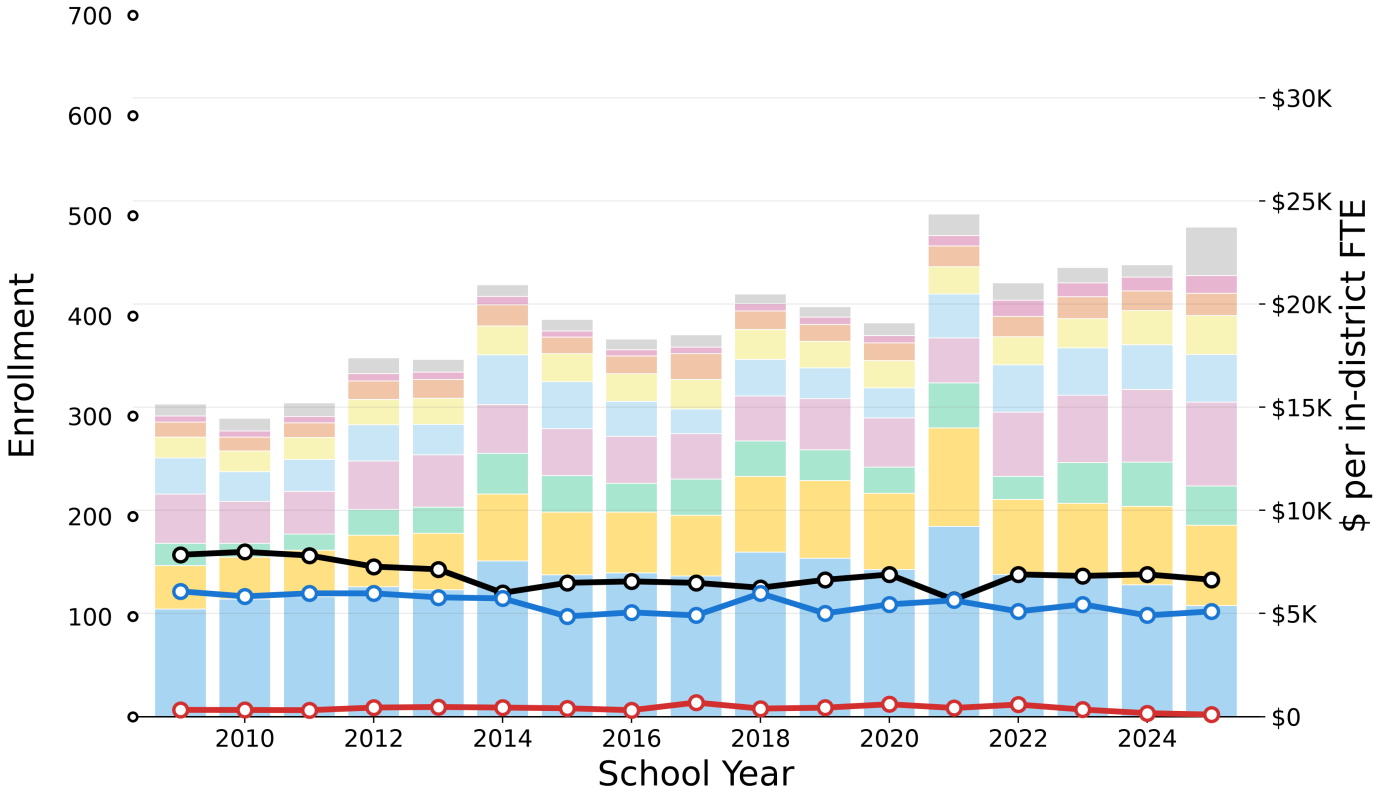


Figure 27

Table 35

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$575	+9.2%	+15.4%	+31.2%	$2,352
Guidance, Counseling and Testing	$310	+6.6%	+11.5%	+19.0%	$865
Administration	$715	+2.6%	+2.8%	+4.6%	$1,071
Instructional Leadership	$1,015	+4.0%	+3.4%	+7.5%	$1,895
Operations and Maintenance	$1,755	+1.7%	+0.2%	+9.6%	$2,307
Other Teaching Services	$2,385	+3.4%	+6.0%	+11.2%	$4,062
Pupil Services	$1,085	+3.6%	+0.6%	+8.2%	$1,900
Insurance, Retirement and Other	$2,098	+3.9%	+2.5%	+1.1%	$3,891
Teachers	$5,218	+0.2%	-2.4%	-5.5%	$5,389
Total	$15,156	+2.8%	+2.1%	+4.5%	$23,732

Table 36

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	162	-1.0%	+0.2%	-0.7%	137
Foundation Enrollment	125	-1.1%	+0.5%	-1.3%	105
Out-of-District FTE Pupils	7	-7.3%	-13.3%	-30.5%	2

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

Above cohort baseline (Small (101-700))

CAGR = (End/Start)^(1/years) – 1

Below cohort baseline (Small (101-700))

→ *Compare to Tiny (0-100 FTE) cohort: PPE and Enrollment (page 20)*

Section 3 — Selected districts

Leverett — Chapter 70 Aid and Net School Spending (per foundation pupil)

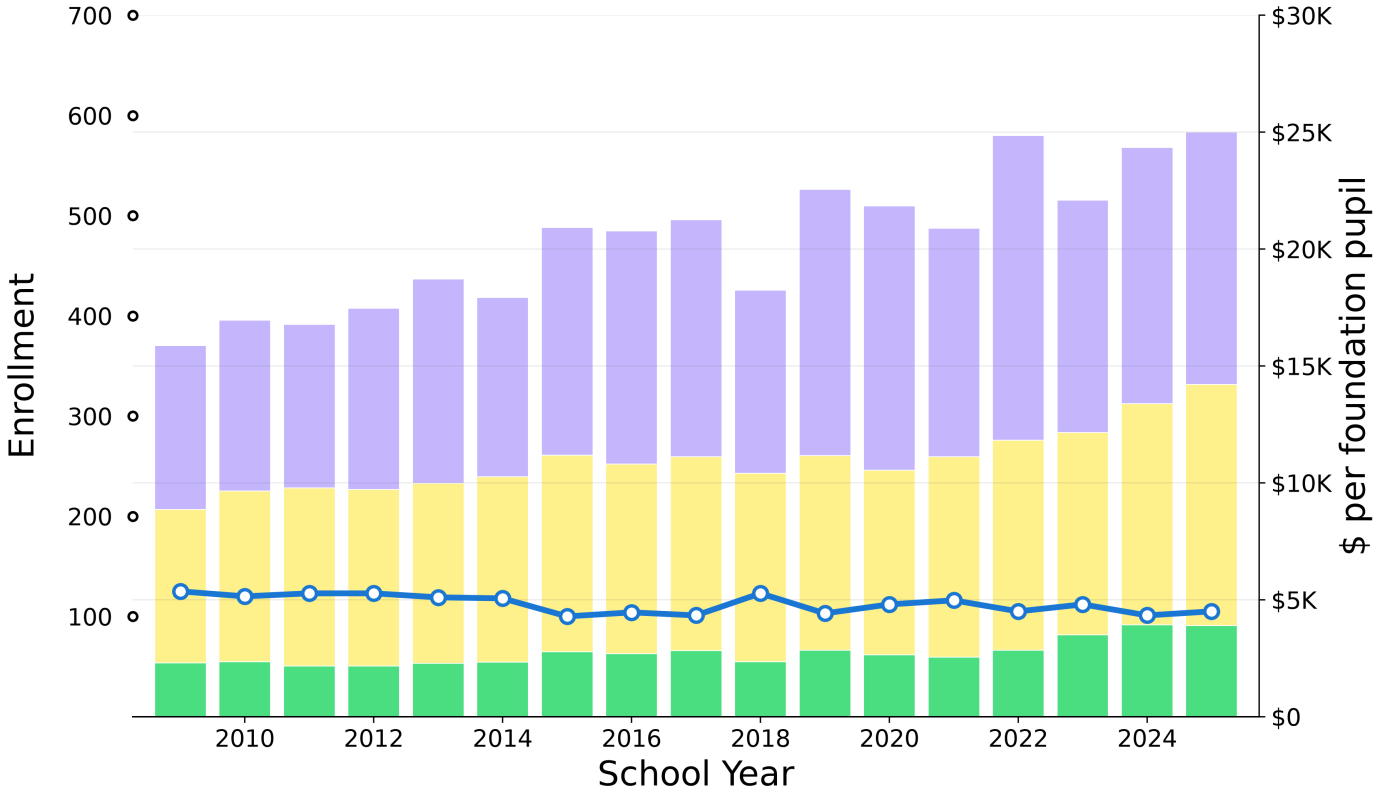


Figure 28

Table 37

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$7,014	2.73%	1.03%	-0.92%	$10,788
Req NSS (minus Ch70)	$6,555	2.88%	2.07%	5.49%	$10,323
Ch70 Aid	$2,303	3.32%	3.43%	8.04%	$3,885
Total (Actual NSS)	$15,872	2.88%	1.80%	2.74%	$24,995

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above cohort baseline (Small (101-700))

Below cohort baseline (Small (101-700))

→ *Compare to Tiny (0-100 FTE) cohort: Chapter 70 Aid and Net School Spending (page 21)*

Section 3 — Specific districts

Pelham — PPE and Enrollment

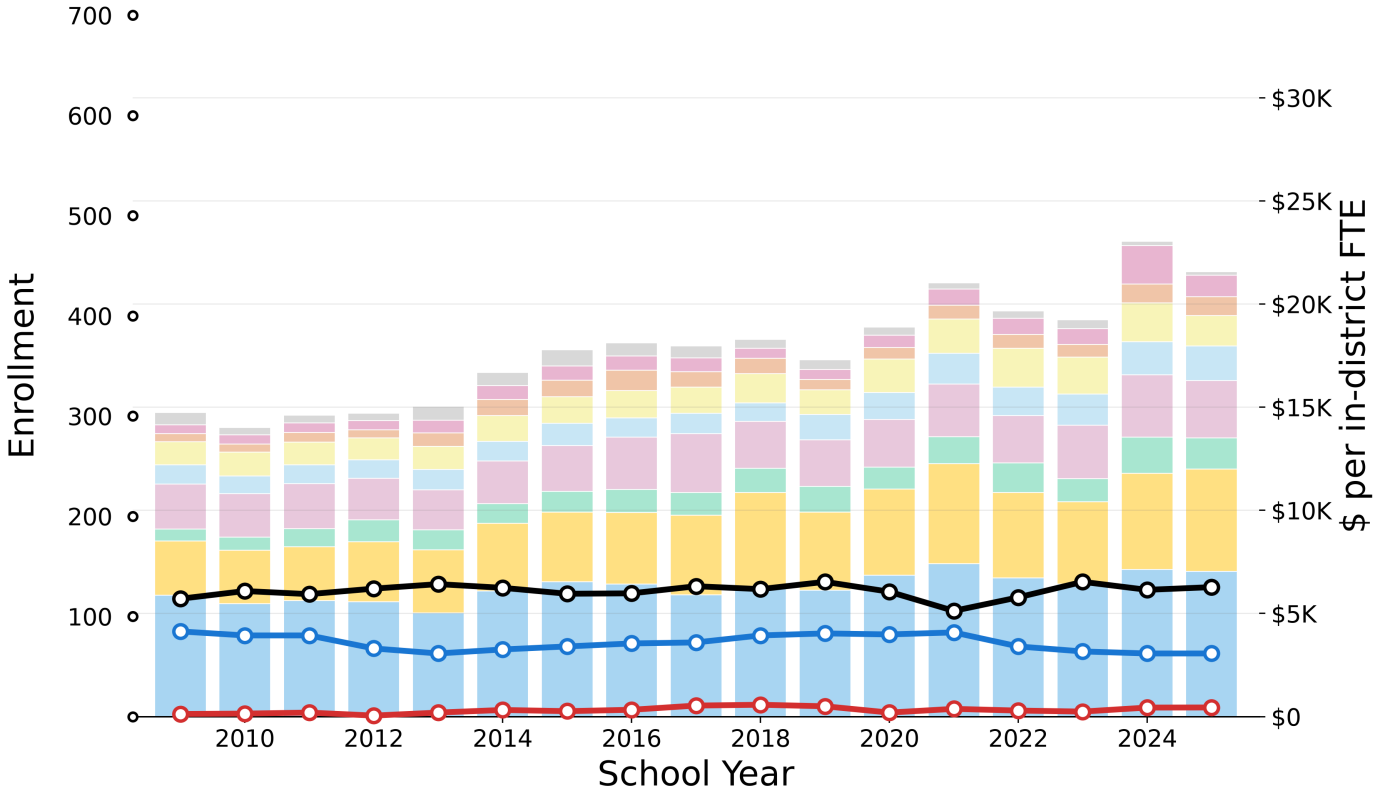


Figure 29

Table 38

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$580	-7.0%	-13.6%	-15.0%	$181
Guidance, Counseling and Testing	$438	+5.6%	+4.1%	+12.4%	$1,044
Administration	$390	+5.4%	+1.2%	+9.5%	$905
Instructional Leadership	$1,123	+1.7%	+1.4%	-1.7%	$1,479
Operations and Maintenance	$920	+3.8%	+4.5%	+5.1%	$1,674
Other Teaching Services	$2,184	+1.5%	+2.3%	+3.7%	$2,786
Pupil Services	$594	+6.0%	+4.2%	+7.2%	$1,503
Insurance, Retirement and Other	$2,615	+4.1%	+3.9%	+3.6%	$4,965
Teachers	$5,889	+1.1%	+0.7%	+0.5%	$7,036
Total	$14,733	+2.4%	+2.0%	+2.7%	$21,573

Table 39

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	118	+0.6%	+0.5%	+0.7%	129
Foundation Enrollment	85	-1.9%	-1.0%	-5.1%	63
Out-of-District FTE Pupils	3	+8.1%	+5.4%	+17.9%	9

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above cohort baseline (Small (101-700))

Below cohort baseline (Small (101-700))

→ *Compare to Tiny (0-100 FTE) cohort: PPE and Enrollment (page 20)*

Section 3 — Selected districts

Pelham — Chapter 70 Aid and Net School Spending (per foundation pupil)

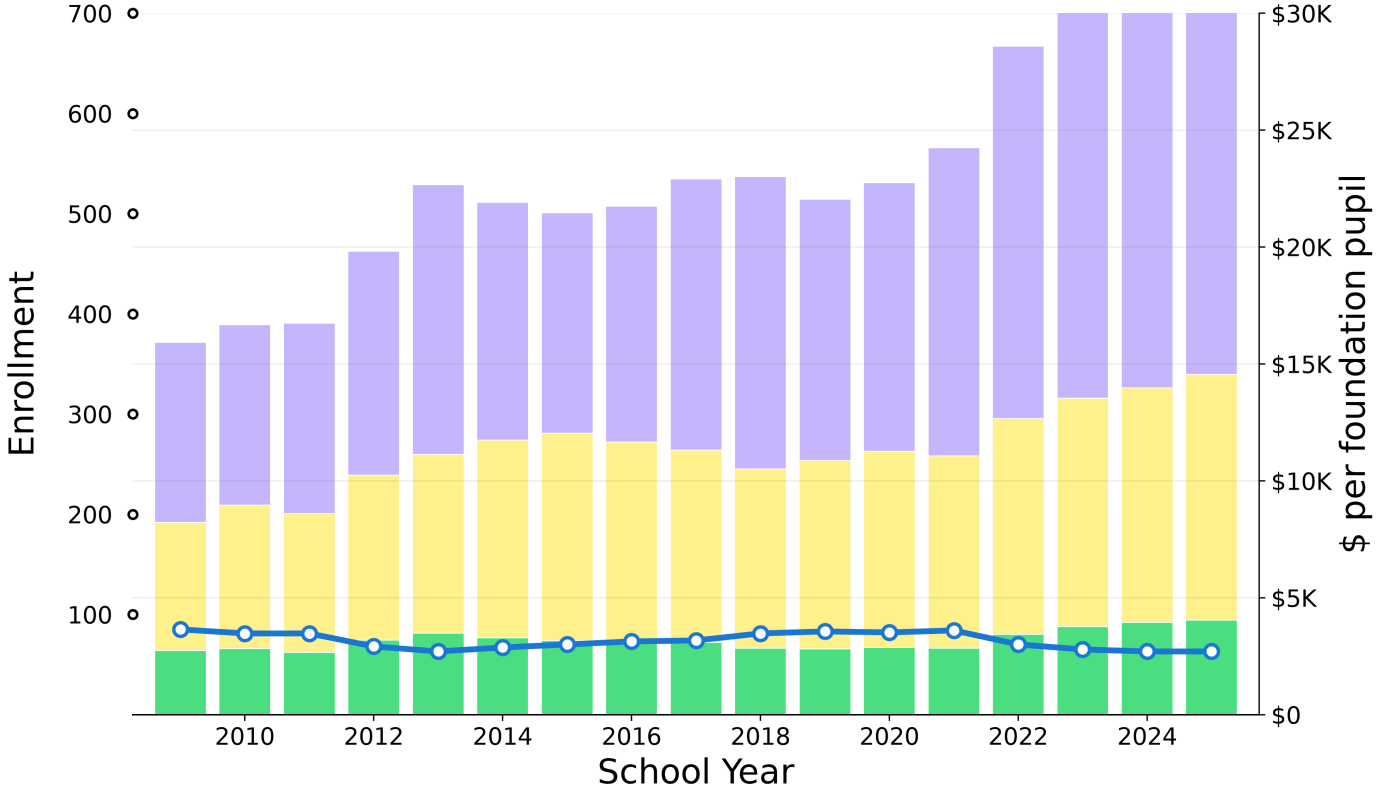


Figure 30

Table 40

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$7,694	6.21%	7.90%	11.92%	$20,171
Req NSS (minus Ch70)	$5,483	4.15%	1.72%	4.61%	$10,505
Ch70 Aid	$2,743	2.46%	2.45%	7.09%	$4,045
Total (Actual NSS)	$15,920	4.99%	4.93%	8.83%	$34,721

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) – 1

Above cohort baseline (Small (101-700))

Below cohort baseline (Small (101-700))

→ *Compare to Tiny (0-100 FTE) cohort: Chapter 70 Aid and Net School Spending (page 21)*

Section 3 — Specific districts

Shutesbury — PPE and Enrollment

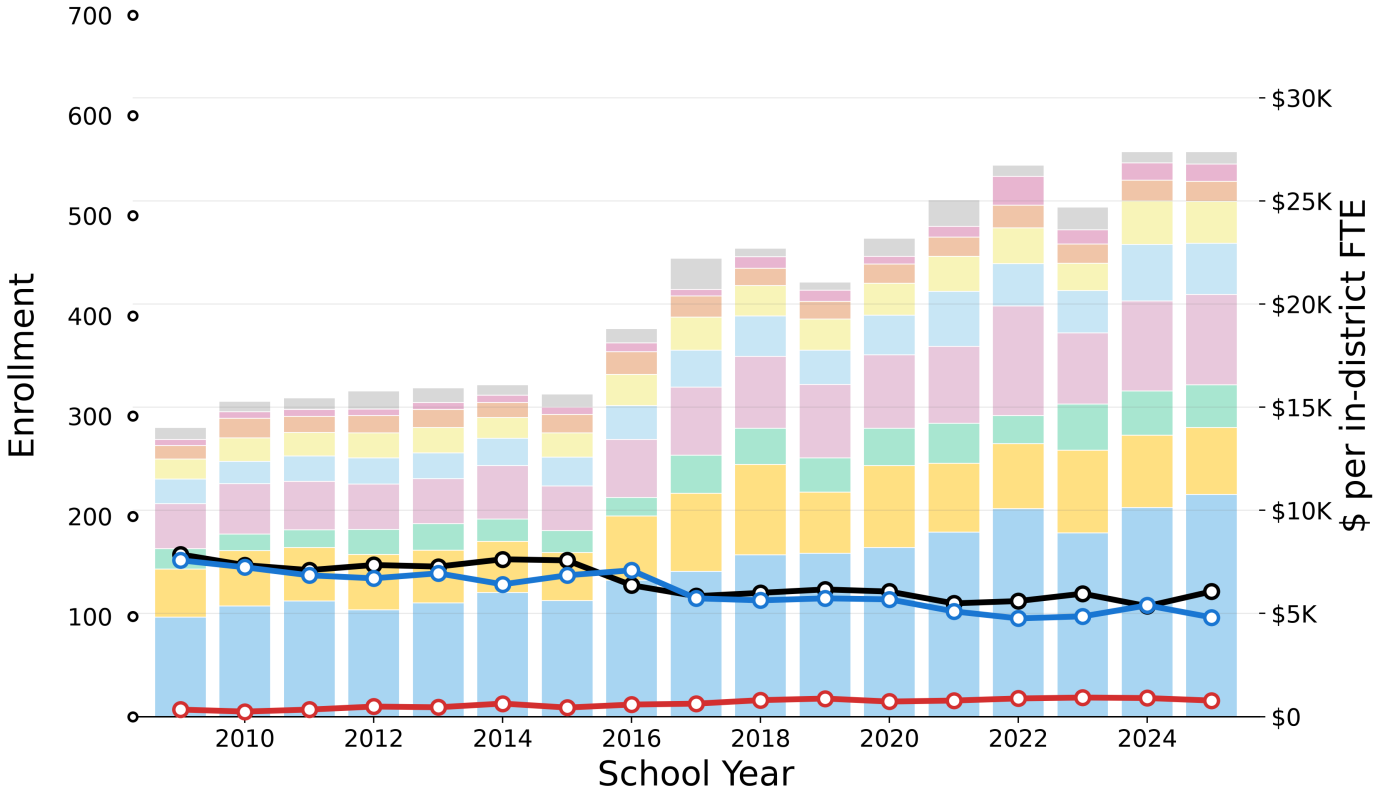


Figure 31

Table 41

Category	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Other	$580	+0.2%	-0.5%	-7.2%	$602
Guidance, Counseling and Testing	$302	+6.7%	+9.1%	+17.2%	$847
Administration	$652	+2.4%	+0.8%	+0.4%	$958
Instructional Leadership	$969	+4.7%	+5.5%	+5.6%	$2,028
Operations and Maintenance	$1,180	+4.8%	+6.1%	+5.4%	$2,493
Other Teaching Services	$2,185	+4.4%	+7.3%	+4.2%	$4,378
Pupil Services	$990	+4.7%	+6.8%	+2.6%	$2,064
Insurance, Retirement and Other	$2,332	+2.1%	+3.4%	-3.9%	$3,248
Teachers	$4,821	+5.2%	+6.7%	+5.6%	$10,771
Total	$14,011	+4.3%	+5.8%	+3.4%	$27,389

Table 42

In-district FTE	2009	CAGR 16y	CAGR 10y	CAGR 5y	2025
In-District FTE Pupils	162	-1.6%	-2.2%	+0.0%	125
Foundation Enrollment	156	-2.8%	-3.5%	-3.3%	99
Out-of-District FTE Pupils	7	+5.3%	+5.9%	+1.3%	16

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔEnrollment| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

Above cohort baseline (Small (101-700))

CAGR = (End/Start)^(1/years) – 1

Below cohort baseline (Small (101-700))

→ *Compare to Tiny (0-100 FTE) cohort: PPE and Enrollment (page 20)*

Section 3 — Selected districts

Shutesbury — Chapter 70 Aid and Net School Spending (per foundation pupil)

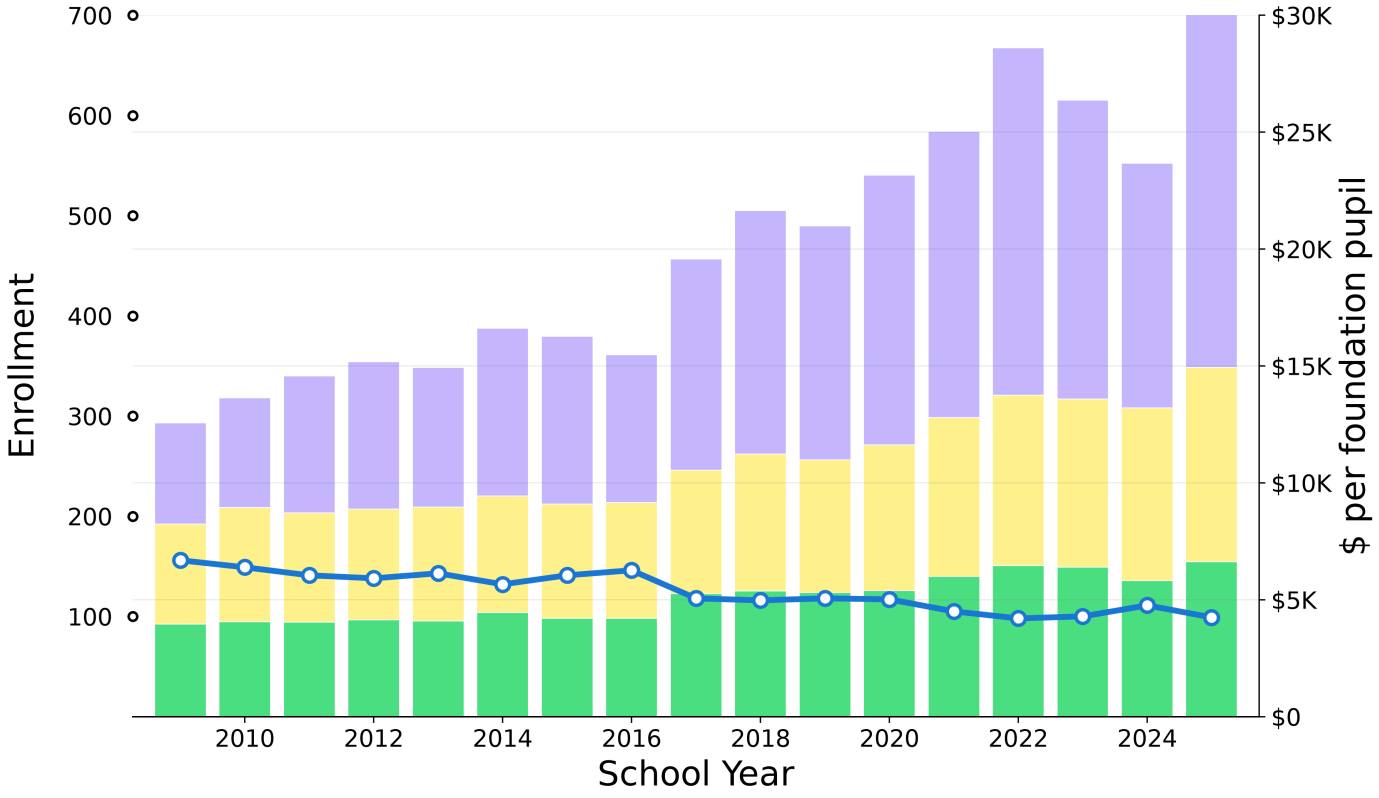


Figure 32

Table 43

Component	2009 $/pupil	CAGR 16y	CAGR 10y	CAGR 5y	2025 $/pupil
Actual NSS (minus Req NSS)	$4,331	8.17%	7.82%	5.71%	$15,220
Req NSS (minus Ch70)	$4,280	4.23%	5.44%	5.88%	$8,307
Ch70 Aid	$3,952	3.29%	4.65%	4.23%	$6,629
Total (Actual NSS)	$12,562	5.63%	6.37%	5.42%	$30,156

Shading vs baseline: |Δ$/pupil| ≥ 5.0%, |ΔCAGR| ≥ 0.5pp

CAGR = (End/Start)^(1/years) − 1

Above cohort baseline (Small (101-700))

Below cohort baseline (Small (101-700))

→ *Compare to Tiny (0-100 FTE) cohort: Chapter 70 Aid and Net School Spending (page 21)*